COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

GROUP – B (ECE, EEE &Mechanical) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1105	Physics	4	3	1	--	4	30	70	100
B16 ENG 1107	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1109	Professional Ethics & Moral Values	2	2	--	--	2	30	70	100
B16 ENG 1111	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1113	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1114	NCC/NSS (Audit)	--	--	--	--	3	--	---	--
Total		**26**	**16**	**4**	**9**	**32**	**280**	**520**	**800**

__Note__: ***Add-On Course: Technical English ****

*** Common to both Group - A and Group - B**

I/IV B.Tech,
Group –B, I-Sem

Code: B16 ENG 1101

ENGLISH

(Common to All Branches)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zukerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

Code: B16 ENG 1102

MATHEMATICS – I
(Common to All Branches)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives , Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period 2C (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler's Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor's Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange's Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz's Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli's Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton's Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy's linear equation – Legendre's Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler's Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval's Formula.

TEXT BOOK:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig,Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

Code: B16 ENG 1103

MATHEMATICS – II
(Common to All Branches)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODE
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouche's Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by 't' – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

Code: B16 ENG 1105

PHYSICS

(Common to ECE, EEE & Mechanical)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS

Thermodynamics

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson - Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India(pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

Code: B16 ENG 1107

ENGINEERING GRAPHICS

(Common to ECE, EEE & Mechanical)

Theory	**: 2 Periods**	**Sessionals**	**: 30**
Lab	**: 3 Periods**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS

Introduction

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

Code: B16 ENG 1109

PROFESSIONAL ETHICS AND MORAL VALUES

(Common to ECE, EEE & Mechanical)

Theory	**: 2 Periods**	**Sessionals**	**: 30**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 70**
		Credits	**: 2**

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

Ethics and Human Values

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

Code: B16 ENG 1111

PHYSICS LAB

(Common to ECE, EEE & Mechanical)

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde's Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton's Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy's constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee's apparatus.
8. Carey Foster's bridge – Verification of laws of resistances.
9. Lee's Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

Code: B16 ENG 1113

WORKSHOP

(Common to ECE, EEE & Mechanical)

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

COURSE OBJECTIVE:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60^0 pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bech vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S. K. and A. K. Choudary, Media Promoters & Publishers Pvt. Ltd.

Code: B16 ENG 1114

NCC/NSS
(ECE, EEE & Mechanical)

Contact Hrs. : 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD -ON COURSE

TECHNICAL ENGLISH

(Common to All Branches)

Theory: 2 Periods

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.

2. Writing Skills
   Business letters, E-mails
   Report Writing
   Writing Proposals & Business Quotations
   Brochures, Newsletters
   How to write an abstract & research paper

3. Speaking Skills
   Technical Descriptions
   PPT Presentations
   Paper Presentations

4. Vocabulary
   Technical Terms

5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

GROUP – B (ECE, EEE &Mechanical) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1203	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1205	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1207	History of Science and Technology	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1210	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1212	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	Sports (Audit)	--	--	--	--	3	--	--	--
	Total	**24**	**14**	**4**	**9**	**30**	**300**	**500**	**800**

Note: ***Add-On Course: Technology Course-I ****

*** Common to both Group-A and Group-B**
 # ECE: B16 EC1208: Electronic Devices and Circuits
EEE: B16 EE1208: Circuit Theory
ME: B16 ME1208: Metallurgy and Materials Engineering

Code: B16 ENG 1201

MATHEMATICS – III

(Common to All Branches)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry
Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1
Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2
Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms
Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval’s identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig,Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

Code: B16 ENG 1203

CHEMISTRY

(Common to ECE, EEE & Mechanical)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinformed plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS

Water Chemistry

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement –Cement concrete -RCC –Decay of concrete.
Refractories: Classification-Properties and Engineering applications.
Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry
Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings-Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics
Definition-Types of polymerization – mechanism of free radical polymerization. plastics-Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants
Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasolilne-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.
LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15^{th} edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

Code: B16 ENG 1205

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS

(Common to ECE, EEE & Mechanical)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction To C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if…else statement, Nesting of if…else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops, One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointes, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt. Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language' B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt. Ltd.
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

Code: B16 ENG 1207

HISTORY OF SCIENCE AND TECHNOLOGY

(Common to ECE, EEE & Mechanical)

Theory	**: 2 periods**	**Sessionals**	**:30**
Exam	**: 3 Hrs.**	**Ext. Marks**	**:70**
		Credits	**: 2**

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS

Historical Perspective of Science and Technology
Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence
Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas
Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** –Effects of a nuclear explosion and India's safety measures**.**

Impact of Science and Technology in Major Areas
Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques-Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development
Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques; **Appropriate Technology** - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East –West Press (P) Ltd., New Delhi.

REFERENCE BOOKS:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

Code: B16 EC 1208

ELECTRONIC DEVICES AND CIRCUITS
(For ECE)

Theory	**: 3 periods**	**Sessionals**	**: 30**
Tutorials	**: 1 period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To give exposure to students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give exposure to students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode.
3. To give exposure to students on rectifier circuits using diodes.
4. To give exposure to students on basics of BJT, JFET and MOSFET and biasing of BJT and FET.
5. To give exposure to students on the analysis of transistor at low and high frequencies.

COURSE OUTCOMES:

After completion of the course the students will be able to
1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJTs and FETs.
2. Use this knowledge to analyze and design basic electronic application circuits.
3. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport Phenomena in SemiConductors
Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

PN junction diode
Open circuited PN junction , PN junction as a rectifier, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode, Point Contact diode, Schottky barrier diode, varactor diode, PIN diode, LED.

Diode Rectifiers
Half wave, Full wave and Bridge Rectifiers with and without filters, Ripple factor and regulation characteristics

Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α,β,γ relation between α and β,γ early effect or basewidth modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.

Field Effect transistors

JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics

Transistor Biasing Circuits

The operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , $V_{BE,}$ & β. Bias compensation, thermal runaway, thermal stability, Biasing of FETs.

Transistors at low and High frequencies

Transistor hybrid model, h-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

TEXT BOOKS:

1. Integrated electronics analog and digital circuits and systems by Jacob Millman, C Halkias, Chetan D Parikh, Tata McGrawhill publications.
2. Electronic Devices and Circuits Theory by Boylestad, Prentice Hall Publications.

REFERENCE BOOKS:

1. Electronic Principles by Albert Malvino, David J Bates, McGrawhill publications.
2. Electronic Devices and Circuits an Introduction by Allen Mottershead, Prenticehall India.
3. Electronic Devices and Circuits by Salivahanan, Tata McGrawhill publications.

Code: B16 EE 1208

CIRCUIT THEORY
(For EEE)

Theory	**: 3 periods**	**Sessionals**	**: 30**
Tutorials	**: 1 period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

COURSE OBJECTIVES:

1. To enlighten and enrich the students to acquire knowledge about the basics of circuit analysis, concepts of AC circuits, DC/AC network theorems, coupled & three phase circuits.
2. To inculcate problem solving skills and understanding of circuit theory through the application of techniques and principles of Electrical Circuits to common circuit problems.

COURSE OUTCOMES:

1. Able to develop an understanding of the basic fundamental electrical laws, elements of electric Networks and learn the techniques to measure voltage and current.
2. Develops the ability to apply circuit theorems to DC and AC circuits.
3. Able to analyse the coupled & three phase circuits.

SYLLABUS

Fundamentals of Electric Circuits
Concepts of Electric circuit: EMF, Current, Potential difference, Power and Energy; Concepts of Network: Active and Passive elements, classification of Linear, Non-linear, Unilateral, Bilateral Lumped and Distributed elements; Reference directions for current and voltage; Voltage and Current sources; Voltage-Current Relations of R, L, C elements; Voltage and Current division; Series and Parallel combinations of Resistance, Inductance and Capacitance

D.C Circuits
Kirchhoff's Laws; Nodal Analysis, Mesh Analysis, Source Transformation, Linearity and Superposition Theorem, Thevinin's And Norton's Theorems, Reciprocity theorem, Maximum Power Transfer Theorem, Star-Delta Transformation.

AC Circuits
The Sinusoidal Forcing Function, Phasor Concept, Average and Effective Values of Voltage and Current, Instantaneous and Average Power, Complex Power, Steady State Analysis using Mesh and Nodal Analysis, Resonance.

Three Phase Circuits

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits, Measurement of Power in Three Phase Power Circuits.

Magnetic Circuits

Magneto motive force(MMF), Reluctance, Magnetic flux; Analysis of magnetic circuit, Analogy between Electric & Magnetic circuits, Series Magnetic circuits, Magnetic leakage, B-H curve, Faraday's Laws of Electromagnetic Induction, Induced EMF, Dynamically Induced EMF, Statically Induced EMF, Self-Inductance, Mutual Inductance.(simple numerical problems)

TEXT BOOKS:

1. Engineering Circuit Analysis By W.H. Hayt Jr & J.E. Kemmerly, 5^{th} Ed., Mc.Graw Hill.

REFERENCE BOOKS:

1. Basic Electrical Engineering By V.K Mehta & Rohit Mehta- S.Chand Pub.
2. Sudhakar & Syam Mohan " Network Analysis", Mcgraw Hill

Code: B16 ME 1208

METALLURGY AND MATERIALS ENGINEERING
(For Mechanical Engineering)

Theory	**: 3 periods**	**Sessionals**	**: 30**
Tutorials	**: 1 period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

PREREQUISITES:

Basic knowledge in solid state chemistry, alloy systems.

COURSE OBJECTIVES:

The overall objective of the course is to impart knowledge about the engineering materials and their properties and predict their behavior under different working conditions and methods to enhance the mechanical properties of the material Specific objectives include:

1. This course will help in learning crystal structures, crystal defects, space lattices.
2. This course will be helps students to understand systems of solid solutions and their nature.
3. To acquaint the knowledge about the cooling curves and phase diagrams of different alloy systems, calculation of weight percent of phases in alloy system by applying Lever rule.
4. To acquaint knowledge about ferrous alloys, particularly Steel and Cast Irons.
5. To impart knowledge about Equilibrium and Non-Equilibrium diagrams.
6. To impart knowledge about different heat treatment and surface hardening methods in improving the mechanical properties of steels.
7. To impart knowledge about the plastic deformation of materials.
8. To impart knowledge about composite materials.

COURSE OUTCOMES:

Upon completion of this course, students should be able to:

1. Understand crystalline solids and their atomic structures.
2. Suggest and recommend necessary engineering materials for specific applications keeping in view of the cost, design, reliability, life, working conditions and properties of the products.
3. Understand different phase transformations in Iron-Iron Carbide diagram and distinguish between steels and cast irons.
4. Select different materials for tools and components based on functional requirements.
5. Use composite materials for different engineering applications like aerospace, automobile, ship building industry, sports item etc.

SYLLABUS

Structure of crystalline solids

Atomic structure & bonding in solids- Crystal structures-calculations of radius, Coordination Number and Atomic Packing Factor for different cubic structures - Imperfection in solids, point defects, Linear defects, Planar defects and Volume defects- Concept of Slip & twinning.

Phase diagrams

Basic terms- phase rule- Lever rule & free energy of phase mixtures coolingcurves- Phase diagram & phase transformation - construction of phase diagrams- binary phase diagrams - Brass, Bronze, Al-Cu and AlSi phase diagrams- Invariant reactions, eutectic, ,peritectic, eutectoid, peritectoid, metatectic&monotectic reactions, Iron carbon phase diagram & microstructures of plain carbon steel & cast iron

Heat treatment

Heat treatment of steel- Annealing, and its types, normalizing, hardening,tempering, martemering, austempering - TTT diagrams, drawing of TTT diagram, TTT diagram for hypo-& hypereutectoid steels, effect of alloying elements, CCT diagram- Martensitic transformation, nature of martensitic transformation- Surface hardening processes like case hardening, carburizing, cyaniding, nitriding Induction hardening, hardenabilty, Jominy end-quench test, Age hardening of Al & Cu alloys Precipitation Hardening

Engineering Alloys

Properties, composition, microstructure and uses of low carbon, mildmedium & high carbon steels. stainless steels, high speed steels, Hadfield steels, tool steels - Cast irons, gray CI, white CI, malleable CI, SC iron-The light alloys- Al & Mg & Titanium alloys- Copper & its alloys: brasses & bronzes- super alloys, Smart materials- Nano materials.

Composite Materials

Classification of composite materials, dispersion strengthened, particlereinforced and fiber reinforced composite laminates properties of matrix and reinforcement materials and structural applications of different types of composite materials.

TEXT BOOKS:

1. "Materials Science & Engineering- An Introduction", William D.Callister Jr. Wiley India Pvt. Ltd. 6th Edition, 2006, New Delhi.
2. Physical Metallurgy, Principles & Practices", V Raghavan.PHI 2nd Edition 2006, New Delhi.

REFERENCE BOOKS:

1. Introduction to Physical Metallurgy by Sidney H Avner,Tata McGraw-Hill Education 1997
2. Materials Science And Engineering: A First Course By V. Raghavan Phi 5th Edition 2011, New Delhi

Code: B16 ENG 1210

CHEMISTRY LAB

(Common to ECE, EEE & Mechanical)

Lab	**: 3 periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl (Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^H Determination by using P^H meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

Code: B16 ENG 1212

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB

(Common to ECE, EEE & Mechanical)

		Sessionals	**: 50**
Lab	**: 3 Periods**	**Ext. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stroed in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that your are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{rd}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6^{th} Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd

Code: B16 ENG 1213

ENGLISH LANGUAGE LAB

(Common to All Branches)

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1 Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012
2 Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3 A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

Code: B16 ENG 1214

SPORTS

(Common to ECE, EEE & Mechanical)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE

TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of course students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in ‘C’ Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler

ESTD: 1980

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
Recognised as Scientific and Industrial Research Organisation
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

ELECTRICAL AND ELECTRONICS ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 2101	Mathematics-IV	4	3	1	--	4	30	70	100
B16 EE 2101	Network Analysis & Synthesis	4	3	1	--	4	30	70	100
B16 EE 2102	Electro Magnetic Field Theory	4	3	1	--	4	30	70	100
B16 EE 2103	Electrical Measurements & Instruments	4	3	1	--	4	30	70	100
B16 EC 2104	Electronics Devices &Circuits	4	3	1	--	4	30	70	100
B16 ME 2106	Engineering Mechanics &Strength Of Materials	4	3	1	--	4	30	70	100
B16 EE 2106	Networks & Measurements Lab	2	--	--	3	3	50	50	100
B16 EC 2105	Electronics Devices &Circuits Lab	2	--	--	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 ENG 2106	Industry Oriented Training.	1	--	--	2	2	50	--	50
TOTAL		**31**	**19**	**7**	**8**	**34**	**380**	**570**	**950**

II/IV B.Tech-I Sem
EEE

Code: B16 ENG 2101

MATHEMATICS – IV
(Common to CIV,ECE,EEE & ME)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students learn

1. The concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Green's, Stokes' and Divergence theorems
3. Classification of 2^{nd} order Partial Differential Equations as well as solution of 1-Dimensional Wave equation and 1-Dimensional Heat equation
4. the concept of Analytic function, CR equations
5. Cauchy's Integral Theorem and Integral Formula
6. Taylor and Laurent series, Residues and Residue theorem

Course Outcomes:

Students will be able to

1. Apply the concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Determine scalar potential, circulation and work done
3. Evaluate integrals using Green's, Stokes' and Divergence theorems
4. Obtain the solution of 1-D wave equation and 1-D heat equation
5. Determine the zeroes and poles of functions and residues at poles
6. Evaluate certain real definite integrals that arise in applications by the use of Residue theorem

SYLLABUS

Vector Calculus-1

Definitions of Scalar and Vector point functions, Differentiation of vectors, Vector differential operator del, Del applied to scalar point function – gradient, Del applied to vector point function- divergence and curl, physical interpretation of gradient, divergence and curl(without proof), Del applied twice to a point function, Del applied to product of two functions, Irrotational and Solenoidal Fields, scalar potential

Vector Calculus-2

Integration of vectors, line integral, circulation, work done, surface integral, Flux, Green's, Stokes' and Gauss Divergence Theorems (Without proofs). Introduction to orthogonal curvilinear coordinates, cylindrical polar coordinates and spherical polar coordinates.

Applications Of Partial Differential Equations

Classification of second order partial differential equations, Method of separation of variables, One –dimensional wave equation- vibrations of a stretched string (no derivation)-, one-dimensional heat equation – Heat flow along a long horizontal bar (no derivation) (problems on heat equation involving homogeneous end conditions only), two dimensional Laplace equation in Cartesian coordinates.

Complex Variables-1

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n, e^z, sin z, sinh z and log z **.**

Limit and continuity of a function of the complex variable, derivative, analytic function, properties of Analytic functions, Cauchy- Riemann equations, Harmonic functions and Orthogonal system, application of analytic function to flow problems, geometric representation of w=f(z), conformal mapping – Bilinear transformation only.

Complex Variables-2

Integration of complex functions, Cauchy's theorem, Cauchy's integral formula (statements only) . Taylor and Laurent series expansions of functions (statement of theorems only), zeros and singularities, Residue, calculation of residues, Cauchy's Residue theorem (without proof), Evaluation of real and definite integrals- integration around a unit circle

Text Book:

1. "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

Code: B16 EE 2101

NETWORK ANALYSIS & SYNTHESIS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To learn source free and forced responses of RL, RC and RLC circuits and evaluation of initial conditions
2. To learn the concept of two-port network analysis and find models using different parameter sets.
3. To learn the concepts of Laplace transform and its application to circuit analysis.
4. Fundamental understanding of the mathematics used to analyze, evaluate, and design transmission network problems.
5. Understand methods for designing transmission networks using synthesis by pole zero methods, foster and cauer form methods.

Course Outcomes:

CO1:Students will outline the significance of energy storing elements (Inductance & Capacitance) in circuits and study transient behavior of responses.
CO2:Students will learn to apply Laplace transform technique for circuit analysis and know its advantages.
CO3:Students will learn to apply two-port network analysis for devices like amplifiers, transmission lines and understand how magnetic coupling can be included in circuit models.
CO4: Students will learn the concept of network functions, poles and zeros and to determine the response of network from poles and zeros.
CO5: Students will learn to apply the synthesis procedure for RC, LC &RL networks (Foster, Cauer methods).

SYLLABUS

DC Transients:
Inductor, capacitor, source free RL, RC & RLC response, evaluation of initial conditions, application of unit-step function to RL, RC & RLC circuits, concepts of natural, forced and complete response.

Laplace Transform Techniques:
Transforms of typical signals, response of simple circuits to unit step, ramp and impulse functions, initial and final value theorems, convolution integral, time shift and periodic functions, transfer function.

Coupled Circuits & Two-port Network parameters:
Magnetically coupled circuits, dot convention, reciprocity theorems, concept of duality,
Two-port Network parameters - Z, Y, H & T parameters.

Network Functions:
Generalized network functions(driving point and transfer), Network functions for ladder & T-networks, concept of poles and zeros, determination of free and forced response from poles and zeros.

Network Synthesis:
Synthesis problem formation, Hurwitz polynomials, properties and test for positive real functions, elementary synthesis operations, Foster and Cauer Forms of LC, RC and RL networks.

Text Books

1. Engineering circuit analysis by W.H. Hayt Jr & J.E. Kemmerly, McGraw Hill Education; Eighth edition (4 August 2013).

Reference Books:

1. Network analysis by M.E. Van Valkenberg, 3rd Edition,2006, Prentice Hall India Learning Private Limited.
2. Modern network synthesis by M.E. Van Valkenberg, John Wiley & Sons ,1966.

Code: B16 EE 2102

ELECTRO MAGNETIC FIELD THEORY

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. All the electric equipment is developed by using the magnetic material, conductors and insulators. It is very much essential to know the behavior of these materials in the presence of electric and magnetic fields.
2. The main objective of this course is to provide the basic concepts about the effects of electric and magnetic fields on conductors, magnetic materials, and insulators under various operating conditions.

Course Outcomes:

Students are able to

CO1. Find the electrostatic and magneto static fields for different configurations.

CO2. Apply various principles and laws to estimate the effect of electric and magnetic fields.

CO3 Distinguish between the effects of electrostatic and magneto static fields.

CO4. Apply Maxwell's equations for static and time varying fields.

CO5. Analyze the EM wave in different domains and compute average power density

SYLLABUS

Coordinate systems:

Rectangular, cylindrical and spherical coordinate systems.

Electrostatics:

Coulomb's law and superposition principle, different types of charge configurations, electric flux, electric field intensity and electric flux density, electric field intensity and electric flux density due to different charge configurations, Gauss's law in integral form and point form in terms of D, applications of Gauss' law, Divergence theorem.

Electric potential, calculation of electric potential for giving charge configuration, electrostatic energy, Electrostatic boundary conditions, basic properties of conductors in electrostatic fields, capacitance, Poisons and Laplace's equations, solutions of Laplace's equations, uniqueness theorems, methods of images, electric dipoles, polarization of dielectrics, bound charges.

Magneto statics:

Biot-savart's law, determination of magnetic field intensity and magnetic flux density due to various steady current configurations, continuity equation, curl of $\overline{H}$, Ampere's circuital law in integral and differential form, applications of Ampere's law, Stokes theorem.

The scalar and vector magnetic potential and calculation of magnetic field through the vector magnetic potential for given steady current configurations, magnetostatic boundary conditions.

The magnetic dipole, magnetization, properties of magnetic materials, torques and forces on magnetic dipoles, bound current, Faraday's laws, Lenz's law, inductance and energy in magnetic fields.

Time varying fields and Maxwell's equations:
Lorentz force equation, Maxwell's equations, modification of ampere's circuital law for time varying fields – displacement current and current density, the uniform plane wave, plane wave propagation, phase velocity and wavelength, intrinsic impedance, perfect dielectrics, attenuation, phase and propagation constants, skin depth, the poynting vector, poynting theorem and power considerations.

Textbooks:

1. Introduction to electro dynamics by D.J. Grifiths, PHI Learning; 3rd Edition (2012).
2. Engineering electromagnetics by William H. Hayt , John A. Buck McGraw-Hill Publishing Co. (2001).

Reference books:

1. Principles of Electromagnetics by Mathew N.O. Sadiku, Oxford; Fourth edition. (2009).

Code: B16 EE 2103

ELECTRICAL MEASUREMENTS & INSTRUMENTS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course objectives

1. The students learn about measuring instruments to measure electrical quantities like current, voltage etc.
2. This course familiarizes with Wattmeter's, Energy meters, power factor meters, frequency meters etc.
3. The students learn to measure resistance, inductance, capacitance, etc. using bridges
4. The course familiarizes ballistic galvanometer, flux meter, testing of ring and bar specimens for magnetic measurements.
5. Students learn about AC & DC potentiometers, phase & amplitude measurements, use of a CRO, instrument transformers.

Course Outcomes

Upon successful completion of this course, the students will be able to:

CO1: illustrate the characteristics of measuring instruments (K3)
CO2: Discriminate measuring instruments based on their principle & operation(K4)
CO3: Calculate power and energy in 1$\emptyset$, 3$\emptyset$ & polyphase circuits (K3)
CO4: Measure electrical parameters using a bridge (K3)
CO5: Find magnetic measurements using Ballistic Galvanometers and Flux meters. (K4)
CO6: Apply potentiometers & instrument transformers to measure electrical elements, calibration of the meters. (K3)

SYLLABUS

Philosophy of measurement
Methods of measurement, measurement system, classification of instrument system, characteristics of instruments & measurement system, errors in measurement & its analysis, standards.

Analog measurement of electrical quantities
Moving coil, moving iron, Electrodynamometer type, electrostatic and induction type instruments, electrodynamic wattmeter, three phase wattmeter, power in three phase system, errors & remedies in wattmeter and energy meter. Extension of instrument range, introduction to measurement of frequency and power factor.

Measurement of parameters
Different methods of measuring low, medium and high resistances, measurement of inductance & capacitance with the help of AC bridges. DC potentiometers and its applications. AC potentiometer - types & applications.

Magnetic measurement
Ballistic galvanometer, flux meter, determination of B-H curve and hysteresis loop, measurement of iron losses, current transformers and potential transformers.

Cathode Ray Oscilloscope:
Basic CRO circuit (block diagram), cathode ray tube (CRT) & its components , application of CRO in measurement of B-H curve.

Digital measurement of electrical quantities
Digital Instruments, Concept of digital measurement,Analog to digital & Digital to analog conversion, advantages of digital Instruments, digital display units, Resolution in digital meters,sensitivity & Accuracy of digital meters.

Text Books:

1. E.W. Golding & F.C. Widdis, "Electrical Measurement &Measuring Instrument", Reem Publications Pvt. Ltd.; Third edition (2011).
2. A.K. Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Co. (P) Limited; 2014 edition (2015)
3. W.D. Cooper," Electronic Instrument & Measurement Technique " Prentice Hall International.

Reference Books:

1. Forest K. Harries,"Electrical Measurement",Willey Eastern Pvt. Ltd. India .
2. M.B. Stout,"Basic Electrical Measurement" Prentice Hall of India.
3. Rajendra Prashad ,"Electrical Measurement &Measuring Instrument" Khanna Publisher.
4. J.B. Gupta, "Electrical Measurements and Measuring Instruments", S.K. Kataria & Sons, 2012 Edition.

Code: B16 EC 2104

ELECTRONICS DEVICES &CIRCUITS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give the exposure to the students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode etc.
3. To give the exposure to the students on rectifier circuits using diodes.
4. To give the exposure to the students on basics of BJT, JFET and MOSFET and biasing of BJT and FET's.
5. To give the exposure to the students on the analysis of transistor at low and high frequencies.

Course outcomes:

After completion of the course the students will be able to
CO1: Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJT's and FET's.
CO2: Use this knowledge to analyze and design basic electronic application circuits.
CO3: Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport Phenomena in Semiconductors

Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, the continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

PN junction diode

Open circuited PN junction , PN junction as a rectifier, current components in a PN diode,
V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode, Point Contact diode, Schottky barrier diode, varactor diode, PIN diode, LED.

Diode Rectifiers

Half wave, full wave and bridge rectifiers with and without filters, ripple factor and regulation characteristics.

Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α,β,γ relation between α and β,γ early effect or basewidth modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, comparison of CE, CB and CC configurations, break- down in transistors, photo transistor.

Field Effect transistors
JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics.

Transistor Biasing Circuits
The operating point, bias stability, different types of biasing techniques, stabilization against variation in I_{co}, $V_{BE,}$ & β. bias compensation, thermal runaway, thermal stability, biasing of FETs.

Transistors at low and High frequencies
Transistor hybrid model, H-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, high frequency model of transistor.

Text Books:

1. Integrated electronics analog and digital circuits and systems: Jacob Millman, C Halkias, Chetan D Parikh.
2. Electronic Devices and Circuits Theory, BoyIsted, Prentice Hall Publications.

Reference Books:

1. Electronic Devices and Circuits by G.S.N.Raju., Published by I.K. International 2006, pbk, 2006.
2. Electronic Devices and Circuits by Salivahanan., 2nd Edition, Tata McGraw-Hill pub.

Code: B16 ME 2106

ENGINEERING MECHANICS & STRENGTH OF MATERIALS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To make the students to understand the principles of the effect of forces under the static and dynamic conditions and apply them to some practical applications.
2. To make the students to understand the principles of the effect of forces on deformable rigid bodies under various loading conditions, and thus measure various types' stresses such as direct stresses, bending stresses, torsional stresses

Course Outcomes:

Students will be able to:

CO1: Evaluate the forces in concurrent and coplanar force systems, using various principles and also under different conditions of equilibrium. Analyze the forces in various applications and apply the concepts of friction to some basic applications of Electrical engineering.

CO2: Understand and apply principles of parallel force systems to find centroid and moment of inertia of different objects.

CO3: Apply the concepts of kinematics and kinetics to analyze force on particles under rectilinear.

CO4: Distinguish between various mechanical properties like yield strength, ultimate strength etc., of a given material and also to determine various types of direct stresses. Analyze the effect of shear force & bending moment on various beams.

CO5: Determine the bending stresses in different beams of various cross sections and to find torsional stresses in shafts

SYLLABUS

Part –A: Engineering Mechanics

Statics:

Fundamentals of Mechanics: Basic Concepts, Force Systems and Equilibrium, Moment and Couple, Principle of Superposition & Transmissibility, Varignon's theorem, Resultant of force system – Concurrent and non concurrent coplanar forces, Condition of static equilibrium for coplanar force system, concept of free body diagram, applications in solving the problems on static equilibrium of bodies.

Friction Concept of dry friction, limiting friction, angle of friction, Friction problems related to connecting bodies and ladder.

Properties of bodies:

Center of Gravity: Center of Gravity of Plane figures, Composite Sections and shaded areas.

Area Moment of Inertia: Parallel and Perpendicular axis theorem, Moment of Inertia of symmetrical and unsymmetrical sections

Dynamics:
Kinematics – Introduction to kinematics, Equations of motion for uniform and variable motion; Projectiles.
Kinetics – D'Alemberts principle, Work energy method, Impulse momentum methods.

Part – B: Strength of Materials

Simple Stresses and Strains: Stresses and Strains, stress-strain curve, Bars of uniform, varying and tapered cross –sections, Poisons ratio, volumetric strain and relation between moduli of elasticity
Shear Force and Bending Moment: Cantilever, Simply Supported and Overhanging beams subjected to point loads and uniformly distributed loads.

Bending stresses in beams: Theory of pure bending, Flexure formula, Section modulus for cantilever and simply supported beams having symmetrical and unsymmetrical sections
Torsion of Shafts: Torsion equation for circular shaft, polar modulus and related problems.

Text Books:

1. Engineering mechanics by Bhavikatti. New age international.
2. Engineering mechanics by A.K. Tayal.
3. S. Ramamrutham & R, Narayanan, Strength of Materials, Dhanpat Rai publications.
4. R.K. Bansal "A Text Book of Strength of Materials, Lakshmi Publications Pvt. Ltd,NewDelhi

Reference Books:

1. Engineering Mechanics by S.Timoshenko and D.H. Young McGraw-Hill.
2. Mechanics of Materials by E P Popov
3. Dr Sadhu Singh, Strength of Materilas.

Code: B16 EE 2106

NETWORKS & MEASUREMENTS LAB

		Sessionals	**: 50**
Lab	**: 3 Period**	**Ext. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives:

1. To learn to make simple electric circuits by using different sources, loads and components and verify basic laws.
2. To experimentally verify various theorems of circuit analysis.
3. To learn to find circuit models for two-terminal devices and two-port networks.
4. To explore the sinusoidal steady state behavior and resonance phenomenon in electric circuits.
5. To measure different electrical parameters by using different meters.

Course Outcomes:

CO1: Students will gain the skill to make and experiment with practical electric circuits.
CO2: Students will be able to measure voltage, current, power in practical electric circuits.
CO3: Students will know the significance of various theorems and their applications.
CO4: Students will be able to assess the behavior of electric circuits.
CO5: Students will be able to calibrate single phase energy meter, voltmeter & wattmeter
CO6: Students will be able to measure resistance, inductance & capacitance.

LIST OF EXPERIMENTS

1. Verification of Ohms Law and resistance of a filament Lamp
2. Verification of superposition theorem
3. Verification of Thevinen's theorem
4. Verification of Norton's theorem
5. Verification of maximum power transfer theorem
6. Series resonance
7. Calculation two port network parameters
8. Calibration of wattmeter
9. Calibration of energy meter
10. Three voltmeter method
11. Measurement of 3 phase power using two wattmeter method
12. Parameters of choke coil.
13. Measurement of three phase power by using 2 C.T's and Single Wattmeter
14. Crompton's DC potentiometer
15. Kelvin's double bridge
16. Schering bridge

Reference Books:

1. A.K. Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Co. (P) Limited; 2014 edition (2015).
2. Engineering circuit analysis by W.H. Hayt Jr & J.E. Kemmerly, McGraw Hill Education; Eighth edition (4 August 2013).

Code: B16 EC 2105

ELECTRONIC DEVICES & CIRCUITS LAB
(Common to ECE & EEE)

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To familiarize the student with test and measuring equipment like CROs, Multimeters, Ammeters, Voltmeters etc. and also to prepare the student to use signal generators, bread boards and to make the student identify the terminals of basic electronic devices like diodes, transistors and JFETs.
2. To familiarize the student with features of Multisim and to prepare the student to construct and simulate various electronic circuits using Multisim.
3. To make the student study experimentally the characteristics of basic electronic devices like ordinary pn diodes, LEDs, Zener diodes, BJTS, JFETs and rectifiers with and without filters.
4. To make the student to conduct experiments to analyze various parameters of BJT amplifiers and FET amplifiers.

Course Outcomes:

After the successful completion of the lab course, the students will be able

1. To understand the role of basic electronic devices like ordinary Pn diodes, Zener diodes, LEDs, BJTS and JFETs in achieving various functionalities like rectification, voltage regulation, amplification, switching action etc. in various electronic circuits.
2. To construct and simulate different electronic circuits using Multisim.
3. To have the hardware skills and software skills required in the design of electronic systems for various applications.

List Of Experiments

1. V-I characteristics of semiconductor diode, LED and Zener diode.
2. Half wave and full wave rectifier with and without filters.
3. Input and output characteristics of transistor in CE configuration.
4. Transistor biasing circuits and transistor as a switch.
5. CE amplifier.
6. JFET common source amplifier.

List Of Simulation Experiments

7. V-I characteristics of semiconductor diode, LED and Zener diode.
8. Regulation characteristics of Zener diode.
9. Input and output characteristics of transistor in CB configuration
10. JFET Characteristics.
11. CC amplifier
12. JFET common source amplifier

Reference: Lab Manual

Code: B16 ENG 2104

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	**: 1 Period**	**Sessionals**	**: 50**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to
1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvementand motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts .
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informalcontexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills
PPT
Describing event/place/thing
Picture Description
Extempore
Debate
Telephonic Skills
Analyzing Proverbs

Vocabulary
Affixes
Pairs of Words

Reading Skills
Reading Comprehension
Reading/Summarizing News Paper Artic

Writing Skills
Designing Posters
Essay writing
Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated basing on Presentation Skills – 30M, Project 20M)

Code: B16 ENG 2106

INDUSTRY ORIENTED TRAINING
(Common to ECE & EEE)

Lab : 2 Periods	**Sessionals : 50**
Exam : 3 Hrs.	**Credits : 1**

Course Objectives:

1. Be familiar with basic Data structures.
2. Master the implementation of linear data structures.
3. Master the implementation of non linear data structures.
4. Be familiar with Object Oriented Concepts.

Course Outcomes:

1. Application using implementation of Data structures.
2. Application using implementation of Linear and non linear Data structures in view of industry.
3. Applications using Object Oriented Concepts in view of industry.

Syllabus: Industry Oriented Applications on following topics.

BASIC CONCEPTS

System Life Cycle, Algorithm Specification, Recursive Algorithms, Data Abstraction, Performance Analysis, Space Complexity, Time Complexity, Asymptotic Notation, Comparing Time Complexities

IMPLEMENTATION (Using C)

Arrays
Stacks
Queues
Linked List
Double linked lists
Trees
Graphs

Applications of linear and nonlinear data structures and solving simple to complex problems in perspective of industry requirements.

Basic Concepts of OOP

Procedural Paradigms, Object Oriented Paradigm, OOP Principles and Terminology, OOP benefits, Procedure and Object Oriented programming languages, advantages and disadvantages, creating class, defining objects in C++ and JAVA.

Applications using OOP in solving simple to complex problems in perspective of industry requirements.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest – 25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

ELECTRICAL AND ELECTRONICS ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EE 2201	Electrical Machines-1	4	3	1	--	4	30	70	100
B16 EE 2202	Signals & Systems	4	3	1	--	4	30	70	100
B16 EC 2206	Analog Electronics Circuits	4	3	1	--	4	30	70	100
B16 ME 2204	Primemovers & Pumps	4	3	1	--	4	30	70	100
B16 EE 2203	Electrical Power Generation,Transmission & Distribution	4	3	1	--	4	30	70	100
B16 ENG 2201	Enivronmental Studies	2	3	1	--	4	30	70	100
B16 ME 2207	Thermal Prime Movers Lab	2	--	--	3	3	50	50	100
B16 EC 2208	Analog Electronic Circuits Lab with Simulation	2	--	--	3	3	50	50	100
B16 EE 2205	Industry Oriented Technology Lab	1	--	--	2	2	50	--	50
B16 ENG 2204	Industry Oriented Training	1	--	--	2	2	50	--	50
	Total	**28**	**18**	**6**	**10**	**34**	**380**	**520**	**900**

II/IV B.Tech-II Sem
EEE

Code: B16 EE 2201

ELECTRICAL MACHINES-1

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

The course will enable the students to understand

1. Electro-mechanical energy conversions in D.C. machines and energy transfer in transformers
2. Principle of operation of DC machines and transformers
3. Speed control methods of DC motors and parallel operation, testing of DC machines and transformers.
4. Different types of three phase transformer connections.

Course Outcomes:

Students are able to

1. Identify the concepts of electro mechanical energy conversion. K2
2. Describe the concepts of construction, operating principle, different types of DC machines and transformers, effects on DC machine and parallel operation of DC generators. K2
3. Interpret the characteristics of DC machines. K3
4. Discriminate different types of speed control methods of DC motors. K4
5. Examine the performance of DC machines and transformers by different testing methods. K4
6. Discriminate different types of transformer connections K4

SYLLABUS

Electromechanical energy conversion- Basic principles of energy, force and torque in singly and multiply excited systems.

Transformers- Principle, construction and operation of single phase transformers, phasor diagram, equivalent circuit, voltage regulation, losses and efficiency. Testing- open & short circuit tests, sumpner's test.

Autotransformers- construction, principle, applications and comparison with two winding transformer.

Three phase transformer: Construction, various types of connection and their comparative features. Parallel operation of single phase and three phase transformers. Three phase transformer connections. Scott connection, tap changing transformers- no load and on load tap changing of transformers. Cooling methods of transformers.

D.C. Machines- Working principle, construction and methods of excitation. D.C generators emf equation, armature reaction,commutation. Compensating winding, characteristics of various types of generators, applications. D.C. motors- torque equation, D.C. shunt, series and compound motors – characteristics & applications.

Starting & Speed control- Starting methods and speed control of D.C. shunt and series motors testing of D.C motors - direct and regenerative methods to test D.C. machines. Swinburne's test, field's test and separation of losses.

Text books

1. Kothari.D.P and Nagrath.I.J., "Electrical machines", McGraw Hill Education; 4 edition (2010).
2. Bimbhra.P.S, Electrical Machinery, Khanna Publishers, 2011.
3. Irving L. Kosow, "Electrical Machines & Transformers", Prentice Hall; 2nd Revised edition 1990.

Reference Books

1. Clayton. A.E„„Performance and Design of direct current machines"' CBS; 1ST edition (2004).
2. Mg Say, theory, "Performance & Design of A.C Machines", CBS publishers.
3. Fitgerald, A.E., Charles Kingsely jr. Stephen D.Umans, "electric machinery" McGraw-Hill; 6th edition (2005).
4. Hill Stephen, Chapman.j, "Electric Machinery Fundamentals", McGraw-Hill Higher Education; 4 edition (2004).

Code: B16 EE 2202

SIGNALS & SYSTEMS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives

1. Coverage of continuous and discrete-time signals and systems, their properties and representations and methods those are necessary for the analysis of continuous and discrete-time signals and systems.
2. Knowledge of time-domain representation and analysis concepts as they relate to difference equations, impulse response and convolution, etc.
3. Knowledge of frequency-domain representation and analysis concepts using Fourier Analysis tools, Z-transform
4. Concepts of the sampling process.

Course Outcomes

Upon the completion of the course, students will be able to:

CO1: Characterize and analyze the properties of continuous and discrete time signals and systems. [K2]
CO2: Apply the convolution for continuous time signals and discrete time signals. [K3]
CO3: Evaluate the Fourier Series of periodic signals. [K1]
CO4: Determine the Fourier Transform and Z-Transform of different type's of signals and make use of their Properties. [K1]
CO5: Convert a continuous time signal to the discrete time domain and reconstruct using the sampling theorem. [K2]

SYLLABUS

Classification of Signals & Systems:
Basic continuous time signals, basic discrete time signals transformations of independent variables, classification of systems, properties of linear time – invariant systems.

Linear Time – Invariant (LTI) Systems:
Representation of signals in terms of impulses for discrete time and continuous time signals, convolution sum and convolution integral. systems described by differential and difference equations. Block diagram representation of LTI systems described by differential and difference equations, singularity functions.
Analogy between vectors and signals, orthogonal vector and signal spaces. Approximation of a function by a set of mutually orthogonal functions.

Fourier analysis:
The response of continuous time LTI systems to complex exponentials – the continuous time and discrete time exponential fourier series, convergence of fourier series.

Fourier Transform:
Fourier transform of continuous time and discrete time aperiodic signals and periodic signals. properties of continuous time and discrete time fourier transforms. Frequency response characterized by linear constant coefficient differential and difference equations. first order and second order systems.

Z –transform:
Z–transform of discrete time sequence, region of convergence. relation between Z and fourier transform, properties of z-transforms. inverse z-transform, determination of transfer function and impulse response of an LTI system, poles and zeros and system stability.

Sampling Theorem :
The effect of under-sampling, methods of reconstruction of a signal from samples, discrete time processing of continuous time signals. sampling in frequency domain, sampling of discrete time signals.

Text Books:

1. Signals and Systems, Alan V. Oppenheim, Alan S. Willsky and Ian T. Young, Prentice-Hall; New edition (1984).

Reference Books:

1. Communication Systems, B. P. Lathi., BS PUBLICATION (2001).
2. Signals and Systems, B. P. Lathi., Oxford University Press; 2nd edition (2004).

Code: B16 EC 2206

ANALOG ELECTRONICS CIRCUITS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

The aim of this course is to

1. Understand the concepts of different type of amplifiers and analyze them.
2. Learn the classification of feedback amplifiers and analyze them.
3. Compare the voltage and power amplifiers and analyze them.
4. Understand the principle of oscillator and analyze different types of sinusoidal oscillators.
5. Learn the classification of tuned amplifiers and analyze them.
6. Understand the concept and analyze applications of op-amp.

Course Outcomes:

After the completion of the course, students will be able to:
CO1: Know the equivalent circuit of multistage amplifier and its analysis. [K3]
CO2: Identify the different feedback topologies and analyze them. [K1]
CO3: Explain the principle of oscillator and design different types of sinusoidal oscillators.[K3]
CO4: Define the difference between voltage and power amplifiers and design different classes. [K1, K3]
CO5: Know that Tuned amplifiers amplify a narrow band of frequencies and will also be able to analyze them.[K2, K3]
CO6: Identify that Op-amp not amplifies but also perform different operations and analyze some applications.[K1,K2]

SYLLABUS

Multistage Amplifiers
Transistor at high frequencies, CE short circuit current gain and concept of Gain Bandwidth Product. BJT and FET RC Coupled Amplifiers at low and high frequencies. Frequency Response and calculation of Band Width of Multistage Amplifiers.

Feed Back Amplifiers
Concept of Feed Back Amplifiers - Effect of Negative Feedback on the amplifier characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.

Sinusoidal Oscillators
Condition for oscillations –LC Oscillators – Hartley, Colpitts, Clapp and Tuned Collector Oscillators – Frequency and amp litude Stability of Oscillators – Crystal Oscillators – RC Oscillators -- RC Phase Shift and Weinbridge Oscillators.

Power Amplifiers

Classification of Power Amplifiers – Class A, Class B and Class AB power Amplifiers. Series Fed, Single Ended Transformer Coupled and Push Pull Class A and Class B Power Amplifiers. Cross-over Distortion in Pure Class B Power Amplifier, Class AB Power Amplifier – Complementary Push Pull Amplifier with trickle Bias, Derating Factor – Heat Sinks.

Tuned Voltage Amplifiers

Single Tuned and Stagger Tuned Amplifiers – Analysis – Double Tuned Amplifier – Bandwidth Calculation.

Operational Amplifiers

Concept of Direct Coupled Amplifiers. Ideal Characteristics of an operational Amplifier – Differential Amplifier - Calculation of common mode Rejection ratio – Differential Amplifier supplied with a constant current – Normalized Transfer Characteristics of a differential Amplifier – Applications of OP-Amp as an Inverting and Non-Inverting Amplifier, Integrator, Differentiator Summing and Subtracting Amplifier and Logarithmic Amplifier. Parameters of an Op-Amp, Measurement of OP-Amp Parameters.

Text Books:

1. Integrated Electronics- Millman and Halkias.
2. Electronic devices and circuits - Mottershead
3. Op-amps and Linear Integrated Circuits – Gayakwad

Reference Books:

1. Electronic Devices and Circuits by G.S.N.Raju., Published by I.K. International 2006, pbk, 2006.
2. Electronic Devices and Circuits by Salivahanan., 2nd Edition, Tata McGraw-Hill pub.

Code: B16 ME 2204

PRIMEMOVERS & PUMPS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

The objectives of the course are:

1. To make the students understand the various types of prime movers which can be connected to generators for power production
2. To impart the knowledge of various types of pumps.

Course Outcomes:

After the completion of the course, students are able to

CO1: Understand the concepts of hydrodynamic force of jets on stationary and moving flat inclined and curved vanes.

CO2: Apply the concepts of momentum equation for finding the forces acting on the vanes of the turbines.

CO3: Understand the carnot, otto, Diesel, Rankine, Joule Cycles.

CO4: Apply the otto, Diesel cycles for finding the performance of S.I and C.I engines.

CO5: Understand the working principle of steam turbines and gas turbines.

CO6: Evaluate the performance characteristics of steam and gas turbines.

CO7:Understand the working principle of centrifugal and reciprocating pumps.

CO8: Evaluate the performance characteristics of centrifugal and reciprocating pumps.

SYLLABUS

Basics of Turbo Machinery

Hydrodynamic force of jets on stationary and moving flat inclined and curved vanes, jet striking centrally and at tip, velocity diagrams, work done and efficiency. Hydraulic Turbines: Classification of turbines, impulse and reaction turbines, pelton wheel, Fancis turbine and Kaplan turbine, wok done, efficiency and draft tube theory.

Thermo dynamic cycles

Carnot, otto, Diesel, Rankine, Joule Cycles- Description and representation on P-V diagram. IC Engines: Working of petrol and Diesel Engines- Two stroke and four stroke engines-Comparison

Turbines

Steam Turbines: Classification – Impulse and reaction turbines – principle of operation – simple impulse turbine, velocity compounding, pressure compounding and pressure – velocity- compounding.

Gas turbines: Simple gas turbine plant, principle of working, Ideal and actual cycles – Open and closes cycles.

Pumps

Reciprocating Pumps: Working, Discharge, slip and indicator diagrams Centrifugal Pumps: Classification, working, workdone- manometric head- losses, efficiencies &specific speed.

Text Books:

1. Fluid Mechanics & Hydraulic Machines – by Modi & Seth, PHI Publications.
2. Engineering Thermodynamics – by P K Nag, Tata McGraw-Hill Companies.

Reference Books:

1. Fluid Mechanices & Hydraulic Machines- by R.K.Bansal, Laxmi Publications.
2. Thermodynamics & Heat Engines – by B.Yadav, Central Book Depot, Allahabad.
3. I C Engines – by V Ganeshan, Tata McGraw-Hill Companies.

Code: B16 EE 2203

ELECTRICAL POWER GENERATION, TRANSMISSION & DISTRIBUTION

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives

1. Understand the general arrangement, principles and components & their functions present in thermal, hydro, nuclear and gas power plants.
2. Understand the load curves and different types of tariffs.
3. Know the performance analysis of transmission lines
4. Give emphasis on mechanical design of transmission line cables and insulators.
5. To study the different types of distribution systems.

Course Outcomes

After completion of the course student will be able to

CO1: Explain the power generation from different energy sources.
CO2: Evaluate different tariffs.
CO3: Analyze the various transmission and distribution systems.
CO4: Design overhead transmission systems under various conditions.
CO5: Calculate Inductance & Capacitance of transmission lines.

SYLLABUS

Electric Power Generation& Economic Considerations:
Layout of thermal, hydro, nuclear and gas power plants, brief description of various parts of different power plants. Load curves and associated definitions, load duration curves, different types of tariffs and examples.

Power Supply Systems& Distribution Systems:
Transmission and distribution systems- D.C 2-wire and 3- wire systems, A.C single phase, three phase and 4-wire systems, comparison of copper efficiency. Primary and secondary distribution systems, concentrated & uniformly distributed loads on distributors fed at both ends, ring distributor, voltage drop and power loss calculation, Kelvin's law.

Inductance & Capacitance calculations:
Types of conductors, line parameters, calculation of inductance and capacitance of single and double circuit transmission lines, three phase lines with bundle conductors. Skin effect and proximity effect.

Performance of transmission lines:
Generalized network constants and equivalent circuits of short, medium, long transmission line. Line performance: regulation and efficiency, Ferranti effect.

Overhead Line Insulators:

Types of insulators, potential distribution over a string of suspended insulators, methods of equalizing potential. Corona: phenomenon of corona, corona loss, concept of radio interference.

Mechanical Design of Transmission Lines :

Different types of tower, sag –tension calculations, sag template, string charts.

Text Books:

1. Wadhwa,C.L., " Electric Power Systems", New Age International Private Limited; Sixth edition (2010).
2. Power System Analysis and Design by Dr. B.R Gupta S Chand & Company; 2005.
3. Nagarath,I.J, and Kothari, D.P., "Power System Engineering", McGraw Hill Education; 2 edition (2007).
4. "A Course in Power Sytems" by J.B Gupta, S.K. Kataria & Sons; 2013 edition.
5. " Principles of power systems" by V.k Mehta & Rohit Mehta by S.CHAND Publications, 3rd edition 2005.

Reference Books:

1. Burke James,J., "Power Distribution Engineering; Fundamentals and Applications " Marcel Dekker 1996.
2. Grainger john, J. and Stevenson ,Jr. W.D., "Power System Analysis", McGraw Hill,1994.
3. Harder Edwin,I., "Fundamentals of Energy Production",John Wiely and Sons,1982.
4. Deshpande, M.V., "Elements of Electric Power Station Design",A.H Wheeler and Co.Allahabad,1979.

Code: B16 ENG 2201

ENIVRONMENTAL STUDIES
(Common to ECE, EEE & ME)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a "CONCERN AND RESPECT FOR THE ENVIRONMENT"

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity**,** Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources**: Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR,10th ed, Thomson publishers, 2004

Code: B16 ME 2207

THERMAL PRIME MOVERS LAB

		Sessionals	**: 50**
Lab	**: 3 Periods**	**Ext. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives:

1. To understand the principle and functioning of various IC engines.
2. Ability to understand the working of two stroke and four stroke engines.
3. Acquiring the knowledge of operation of a turbines and pumps.
4. The way of determination of flash and fire points of oil samples.and their importance is acquired.
5. The procedure for determination of viscosities of oil samples can be understood.

Course Outcomes:

Students will be able to:

CO1: Explain the working principle of different types of IC Engines and illustrate the valve timing and port diagrams of an IC engines.

CO2: Determine the viscosities of oil samples, Flash and Fire point values of fuels.

CO3: Perform the load, Morse, Heat balance and economical speed test on IC Engines.

CO4: Discuss the working principle of different types of hydraulic turbines

CO5: Illustrate the working principle of centrifugal and reciprocating pumps

SYLLABUS

1. Drawing of VTD for four-stroke and PTD of two-stroke engines.
2. Determination of flash and fire points
3. Determination of the kinematic and absolute viscosity of the given sample oils.
4. Load test and smoke test on I.C. engines.
5. Morse test on multi-cylinder engine.
6. Heat balance sheet on I.C. engines.
7. Study of multi-cylinder engines and determination of its firing order.
8. Economical speed test on IC engines.
9. Study on impulse and reaction turbines
10. Study on reciprocating and centrifugal pumps

Reference Books:

1. Thermal Engineering, by R. K. Rajput, Lakshmi Publications.
2. Thermal Science and Engineering by D.S. Kumar, S.K. Kataria and Sons.
3. I.C engines by V. Ganesan, Mc Graw Hill Publications.

Code: B16 EC 2208

ANALOG ELECTRONIC CIRCUITS LAB WITH SIMULATION
(Common to ECE & EEE)

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. This laboratory course enables students to get practical experience in design, assembly and evaluation of analog electronic circuits. They will use Multisim to test their electronic designs.

Course Outcomes:

Students will be able to:

1. Acquire a basic knowledge on simple applications of operational amplifier.
2. Observe the amplitude and frequency responses of negative feedback amplifier and two stage RC coupled amplifier.
3. Design and test sinusoidal oscillators.
4. Design and test a power amplifier.
5. Design,construct,and take measurement of the analog electronic circuits tocompare experimental results in the laboratory with theoretical analysis.
6. Use Multisim to test their electronic design.

LIST OF EXPERIMENTS

1. Design of LC Oscillators (Hartley Oscillator,Colpitts Oscillator)
2. Design of RC Oscillators (Wien Bridge Oscillator,RC phase Shift Oscillator)
3. Design of Basic Applications of Operational Amplifier.
4. Frequency response of Two Stage RC Coupled Amplifier.
5. Frequency response ofCurrent Series Feedback Amplifier(with and without feedback)
6. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
7. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.

LIST OF EXPERIMENTS
(Simulation)

8. Design of LC Oscillators (Hartley Oscillator,Colpitts Oscillator)
9. Design of RC Oscillators (Wien Bridge Oscillator,RC phase Shift Oscillator)
10. Design of Basic Applications of Operational Amplifier.
11. Frequency response of Two Stage RC Coupled Amplifier.
12. Frequency response ofCurrent Series Feedback Amplifier(with and without feedback)
13. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
14. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.

Reference : Lab Manuals

Code: B16 EE 2205

INDUSTRIAL ORIENTED TECHNOLOGY LAB

Lab	**: 2 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 1**

Objective:

This lab is designed to provide students to learn the latest technologies so that they will be Industry ready.

List of Projects :

1. Solar based automated irrigation system.
2. Smart key: a high security based door locked system.
3. Obstacle movement based automatic head lighter blinker and motion controller.
4. Sun tracking solar panel.
5. Foot step power generation.
6. Autonomous solar car to avoid road accidents.
7. Helmet operated smart e bike.
8. Cross roads traffic controller suing e-subway system.
9. Voice command page turning robot for physically challenged people.
10. Automatic fire alerting system in trains

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Record/Report-10 Marks, Exam-10 Marks and Attendance-5 Marks)

Code: B16 ENG 2204

INDUSTRY ORIENTED TRAINING
(Common to ECE & EEE)

Lab : 2 Periods	**Sessionals : 50**
Exam : 3 Hrs.	**Credits : 1**

Course Objectives:

1. Be familiar with core JAVA.
2. Master the implementation of Applet programming.
3. Master the implementation of Networking concepts in core JAVA.
4. Be familiar with CORBA , J2EE, RMI concepts..

Course Outcomes:
1. Application using implementation of core JAVA concepts.
2. Application using implementation of AWT, Applets
3. Applications using Networking concepts in view of industry.

Syllabus: Industry Oriented Applications on following topics.

BASIC CONCEPTS

Fundamentals: HTML, OOP Concepts, Comparing JAVA with C & C++,JAVA Programming language Syntax, Variables, Data types, statements and expressions.

Control Statements: If else, for, while, and do while loops, Switch statements.

Arrays & Structures: One Dimensional & Two Dimensional Arrays, Named Structures.

Functions: Parameter Passing, Static Modifier.

IMPLEMENTATION (Using JAVA)
Classes and Interfaces
Threads and multithreaded programming
packages.

Applications of AWT, Applets and Networking concepts and solving simple to complex problems in perspective of industry requirements.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest – 25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)
III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System
ELECTRICAL AND ELECTRONICS ENGINEERING
I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Sessional Marks	Exam Marks	Total Marks
B16 EE 3101	Electrical Machines-II	4	3	1	--	4	30	70	100
B16 EE 3102	Power System Analysis and Stability	4	3	1	--	4	30	70	100
B16 EC 3105	Linear Integrated & Pulse Circuits	4	3	1	--	4	30	70	100
B16 EE 3103	Control Systems	4	3	1	--	4	30	70	100
B16 EE 3104	Digital Electronics& Logic Design	4	3	1	--	4	30	70	100
#ELE-I	Elective-I	4	3	1	--	4	30	70	100
B16 EE 3107	Electrical Machines-I Lab	2	--	--	3	3	50	50	100
B16 EC 3108	Linear Integrated Circuits & Pulse Circuits Lab with Simulation	2	--	--	3	3	50	50	100
B16 ENG 3102	Verbal & Quantitative Aptitude-I	2	5	--	--	5	100	--	100
#M-I	MOOCS-I	2	4	--	--	4	100	--	100
	Total	**32**	**27**	**6**	**6**	**39**	**480**	**520**	**1000**

#ELE-I	B16 EE3105	Computer Architecture and Organization
	B16 CS 3109	Database Management Systems
	B16 EE 3106	Digital Signal Processing
#M-I	B16 ENG 3103	Basic Coding
	B16 EE3108A	Design of Photovoltaic Systems
	B16 EE3108B	Architectural Design of Digital Integrated Circuits
	B16 EE3108C	Recent Advances in Transmission Insulators

Code:B16 EE 3101

ELECTRICAL MACHINES-II

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. Students understand the concept of principle of operation of 3 phase induction motor, equivalent circuit slip torque characteristics, and circle diagram.
2. They can understand the single phase induction motor, double field revolving theory, and equivalent circuit capacitor start motor.
3. To understand the operation and the principle of synchronous machine, voltage regulation, power angle characteristics synchronizing power, synchronizing torque.
4. To understand different starting methods of synchronous motor, effect of damper windings and Hunting.
5. To understand design procedures of synchronous machine and induction machines.

Course Outcomes:

At the end of this course, students will demonstrate the ability to
1. Understand the concepts of continuous time and discrete time systems.
2. Analyse systems in complex frequency domain.
3. Understand sampling theorem and its implications.

SYLLABUS

Basic concepts of Electrical Machines:
Physical arrangement of windings in stator and cylindrical rotor; slots for windings; single turn coil - active portion and overhang; full-pitch coils, concentrated winding, distributed winding, winding axis, generated emf, Air-gap MMF distribution with fixed current through winding - concentrated and distributed, Sinusoidally distributed winding, winding distribution factor

Induction Machines
Construction, Types (squirrel cage and slip-ring), Equivalent circuit. Phasor Diagram, Torque Slip Characteristics, Starting and Maximum Torque. Losses and Efficiency. Circle Diagram. Effect of parameter variation on torque speed characteristics (variation of rotor resistance). Methods of starting, braking and speed control for induction motors. Generator operation. Self-excitation. Doubly-Fed Induction Machines.

Single phase induction motors
Constructional features, double revolving field theory, equivalent circuit, determination of parameters. Split phase starting methods & applications.

Synchronous Generators
Constructional features, Cylindrical rotor machines, Synchronous Generator-circuit model and phasor diagram, armature reaction, synchronous impedance, voltage regulation and estimation of voltage regulation by EMF, MMF and ZPF methods, Salient pole Machine-Two reaction theory, analysis of phasor diagram, power angle characteristics, determination of x_d and x_q, Parallel operation of Alternators-Synchronization and load division.

Synchronous Motors
Operating principle, circuit model, phasor diagram, effect of load, Operating characteristics of synchronous machines, V-curves, starting methods of synchronous motors.

Text Books:

1. Nagrath& Kothari," Electric Machines" TMH
2. PS Bhimbra, "Electrical Machinery", Khanna Publishers.

Reference Books:

1. Fitzgerald & Kingsley, "Electric Machinery" McGraw Hill
2. Alexander S. Langsdorf, "AC Machines", Tata McGraw Hill.
3. MG Say, "Theory Performance and Design of AC Machines" CBS Publisher

Code: B16 EE 3102

POWER SYSTEM ANALYSIS AND STABILITY

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course objectives

1. Students should acquire knowledge for the design and analysis of electrical power grids.
2. Students are able to learn the modeling of single line diagram and formation of Y-bus and Z-bus.
3. To study the calculation of power flow in a power system network using various load flow techniques,
4. Students learn about different types of power system faults and analysis.
5. Students are able to study the stability of system under steady state and transient period.

Course Outcomes

Upon completing this course

1. Student able to understands and can draw single line diagram of the power system.
2. Student understands different load flow techniques.
3. Students are able to model any complex network into simple mathematical Modelling
4. Student able to analyse different types of fault in a power system
5. Student able to understand stability analysis of power system

SYLLABUS

P.U. Representation: Single Line Diagram, Per Unit Quantities, P.U. Impedance of 3-Winding Transformers, P.U. Impedance Diagram of a Power System.

Load Flow Studies: Formulation of Network Matrices, Load Flow Problem, Gauss-Seidel Method, Newton-Raphson Method & Fast Decoupled Method of Solving Load Flow Problem.

Symmetrical Fault Analysis: 3-Phase Short Circuit Currents and Reactance's of a Synchronous Machine, Fault Limiting Reactors.

Symmetrical Components: The Symmetrical Components, Phase Shift in Delta/Star Transformers, 3-Phase Power in terms of Symmetrical Components.
Un-Symmetrical Faults: Various Types Of Faults – LG, LL, LLG on an Unloaded Alternator, Sequence Impedances and Sequence Networks.

Power System Stability: Concepts of Stability (Steady State And Transient), Swing Equation, Equal Area Criterion, Critical Clearing Angle and Time for Transient Stability, Step by Step Method of Solution, Factors Affecting Transient Stability.

Text Books:

1. Power System Engineering by J.G. Nagarath& D.P. Kothari, TMH Publication
2. Elements of Power System Analysis, William D. Stevenson, Jr, McGraw Hill Pub.

Reference Books:

1. Power System Analysis by Hadi Sadat, McGraw Hill Pub.
2. Power System Stability & Control by PrabhaKundur,TMH Publication.

Code: B16 EC 3105

LINEAR INTEGRATED AND PULSE CIRCUITS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To study the basic principles, configurations and practical limitations of op-amp.
2. To understand the various linear and non-linear applications of op-amp;
3. To analyze and design op-amp oscillators, single chip oscillators and multivibrators;
4. To analyze, design and explain the characteristics and applications of active filters, including the switched capacitor filter.
5. To understand the operation of the 555 timer in mono stable and astable modes and its applications.
6. To provide an overview of the working principles, operation and application of the low pass and high pass circuits and functioning of high pass and low pass RC circuits with various input signals(sine wave, step,square wave, ramp and exponential).

Course Outcomes:

This course provides the foundation education in operational amplifier and other linear integrated circuits. After the completion of course, students are able to

1. Discuss the op-amp's basic construction, characteristics, parameter limitations, various configurations and different applications of op-amp.
2. Analyze and design basic op-amp circuits, particularly various linear and non-linear circuits, active filters, signal generators and 555 timers.
3. Design and conduct experiments, on RC low pass and high pass circuits.
4. Design and conduct experiments, on RL low pass and high pass circuits.
5. Design and conduct experiments on different types of multivibrators.

SYLLABUS

Applications of Operational Amplifiers: Basics of Op-Amp, Instrumentation Amplifiers, Voltage to Current and Current to Voltage Converters. Op-amp As a Comparators, Schmitt trigger, Wave form Generators, Sample and Hold Circuits, Rectifiers.

Active Filters and Oscillators: Butterworth type LPF, HPF first and second order filters, Switched Capacitance Filters. Op-Amp Phase Shift, Wein-bridge and Quadrature Oscillator, Analog Multiplexers.

Special ICs: 555 Timers Introduction, Block diagram,555 timer as an astable and Monostable Multivibrator, Three Terminal IC Regulators, Voltage to Frequency and Frequency to Voltage Converters.

Wave Shaping: High pass and Low pass RC circuits, Response of High pass and Low pass RC circuits to step, square inputs. High pass RC circuit as a differentiator, Low pass RC circuit as an integrator. Diode clippers, Clipping at two independent levels, Clamping Operation, Clamping Circuits using Diode with Different Inputs, Clamping Circuit Theorem, Practical Clamping circuits.

Multivibrators: Transistor as a switch, Switching times of a transistor, Design and Analysis of Fixed-bias and self-bias transistor binary, Commutating capacitors, Design and analysis of Collector coupled MonostableMultivibrator, Expression for the gate width and its waveforms. Design and analysis of Collector coupled AstableMultivibrator, Expression for the Time period and its waveforms, The AstableMultivibrator as a voltage to frequency converter.

Text Books:

1. Microelectronics- Jacob Millman.
2. Op-Amps and Linear ICs- RamakanthGayakwad, PHI, 1987.
3. Pulse Digital and Switching Waveforms, J. Millman ,H. Taub, andM.S.PrakashRaoMcGraw-Hill, Second Edition

Reference Books:

1. Linear Integrated Circuits- D. Roy Chowdhury, New Age International(p) Ltd, 2nd Edition,2003.
2. Pulse and Digital Circuits, A. Anand Kumar, PHI, Second Edition, 2005.

Code: B16 EE 3103

CONTROL SYSTEMS
(Common to ECE & EEE)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To learn the modeling of linear systems using transfer functions and obtain transfer functions for physical electrical and mechanical systems.
2. To learn to represent systems using block diagrams and signal flow graphs and derive their transfer functions..
3. To learn the significance of time response and find it for system analysis in transient and steady state.
4. To learn the concept of stability and know different techniques of stability analysis.
5. To learn the concept of frequency response and its application for control system analysis.

Course Outcomes:

1. Students will be able to model electrical and mechanical physical systems by applying laws of physics
2. Students will be able to represent mathematical models of systems using block diagrams & Signal Flow Graphs and derive their transfer functions
3. Students will be able to analyze systems in time domain for transient and steady-state behavior
4. Students will learn the concept of stability and use RH criterion and Root locus methods for stability analysis.
5. Students will learn to obtain frequency response plots of systems and use them for system analysis and stability assessment.

SYLLABUS

Introduction to control systems : Open loop and closed loop systems- Transfer Functions of Linear Systems– Impulse Response of Linear Systems – Mathematical Modeling of Physical Systems – Equations of Electrical Networks – Modeling of Mechanical Systems – Equations of Mechanical Systems, Analogous Systems.

Block Diagrams of Control Systems: Signal Flow Graphs (Simple Problems) – Reduction Techniques for Complex Block Diagrams and Signal Flow Graphs (Simple Examples)- Feedback Characteristics of Control Systems

Time Domain Analysis of Control Systems: Time Response of First and Second Order Systems with Standard Input Signals – Steady State Error Constants – Effect of Derivative and Integral Control on Transient and Steady State Performance of Feedback Control Systems.

Concept of Stability: Routh-Hurwitz Criterion, Relative Stability Analysis, the Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus (Simple Problems to understand theory).

Frequency Domain Analysis of control systems: Bode Plots- Log Magnitude versus Phase Plots- Polar Plots -Correlation between Time and Frequency Responses -Nyquist Stability Criterion -Assessment of Relative Stability -All Pass and Minimum Phase Systems - Constant M and N Circles.

Text Books:

1.I. J. Nagrath and M. Gopal, 'Control Systems Engineering',New Age International Publishers(6th Edition).
2.Benjamin C. Kuo, ' Automatic Control Systems', PHI (5th Edition).

Reference Books:

1.Katsuhiko Ogata, 'Modern Control Engineering', , PHI (4th Edition).
2.Richard C.Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers(8th Edition)

Code: B16 EE 3104

DIGITAL ELECTRONICS AND LOGIC DESIGN

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To study the basic philosophy underlying the various number systems, negative number representation, binary arithmetic, binary codes and error detecting and correcting binary codes.
2. To study the theory of Boolean algebra and to study representation of switching functions using Boolean expressions and their minimization techniques.
3. To study the combinational logic design of various logic and switching devices and their realization.
4. To study the sequential logic circuits design both in synchronous and Asynchronous modes for various complex logic and switching devices, their minimization techniques and their realizations.
5. To study some of the programmable logic devices and their use in realization of switching functions.

Course Outcomes:

1. Students will be aware of theory of Boolean Algebra& the underlying features of various number systems.
2. Students will be able to use the concepts of Boolean Algebra for the analysis and minimization of Boolean expressions.
3. Students will be able to design of various combinational & sequential logic circuits.
4. Students will be able to design various logic gates starting from simple ordinary gates to complex programmable logic devices & arrays.
5. Students will be to design various logic gates using diodes and transistors.

SYLLABUS

Numbering Systems: Digital systems - Binary, Octal, Decimal and Hex numbering systems – Number base Conversions – (n-1)'s and n's complements of the various numbering systems – Binary arithmetic – Various methods to represent signed binary numbers. Binary Codes:BCD, Excess-3 codes – Binary arithmetic using BCD and Excess-3 codes – Gray code – Error detecting codes: parity checking and Hamming code – Error correcting codes: Hamming code

Boolean Algebra and Boolean Functions: Boolean theorems and postulates – Logic gates – Truth table - Boolean functions – Dual of a function – Complement of a function – Canonical and standard forms – Simplification of Boolean functions using Boolean theorems and postulated, Karnaugh map (K-map) with maximum of 4 variables

Combinational Logic Circuits: Boolean function implementation using AND-OR logic, multilevel NAND and multilevel NOR implementation – Transformation of multilevel NAND and NOR circuits to AND-OR diagram – Combinational logic design - Half adder – Full adder – Half subtractor – Full subtractor – Parallel adder – Parallel adder/subtractor –

Carry look ahead adder – BCD adder – Magnitude comparator –code converters,Decoders – Encoders – Demultiplexer – Multiplexer – Logic implementation using Programmable Logic Devices.

Sequential Logic Circuits: Differences between combinational logic and sequential logic – Flip-flops (R-S, J-K, D, T, Master-slave J-K flip-flop) – Truth tables and excitation tables of the flip-flops, Conversions of flip-flops. Digital Counters-Ripple Counter design, Synchronous Counter design with T, D and J.K. Flip-flops. Shift Registers and Operation Modes.

Realization of Logic Gates Using Diodes & Transistors:
AND, OR and NOT Gates using Diodes and Transistors, RTL, DTL, TTL and CML Logic Families and itsComparison.

Text Book:

1. M. Morris Mano, Digital Design, Prentice-Hall of India Pvt. Limited, New Delhi, 2nd Edition. 2000.

Reference Books:

1. ZviKohavi, Switching and Finite Automata Theory, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2nd Edition, 2008.
2. Frederick J. Hill and Gerald R. Peterson, Introduction to Switching Theory and Logic Design, John Wiley & sons, Inc. New York, 3rd edition, 1981.

Code: B16 EE 3105

COMPUTER ARCHITECTURE AND ORGANIZATION
(Elective-I)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To understand the working of various components of computer and their interconnection.
2. To know the hardware implementation for all micro-operations.
3. Knowledge of instruction set and various instruction formats is provided.
4. Detailed study of CPU, stack organization, peripheral devices and memory organization is provided.

Course Outcomes:

Student can able to:
1. Explain register transfer, memory transfer language, computer registers, computer instructions and addressing modes.
2. Differentiate general register organization & stack organization, Reduced instruction set & complex instruction set computer.
3. Asses instruction cycle, instruction formats.
4. Interpret different memory organization & different modes of transfer.

SYLLABUS

Register transfer and micro operations:
Register transfer language, Register transfer, Bus and memory transfers, Arithmetic Micro Operations, Logic micro operations, Shift micro operations, Arithmetic Logic shift unit

Basic computer organization and Design:
Instruction codes, Computer registers, Computer instructions, Timing and control, Instruction cycle, Memory reference instructions, Input-output and interrupt, complete computer description, Design of Basic computer, Design of Accumulator Logic.

Micro Programmed Control:
Control Memory, Address Sequencing, Micro Program Example, Design of control Unit

CPU Organization:
Introduction, General Register Organization, Stack Organization, Addressing modes, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, program Control(Status Bit Conditions, Conditional Branch Instruction, Subroutine call and return using Stack)

Memory Organization and I/O Organization:
Memory hierarchy, Main memory, Associative Memory, Cache Memory-Associative mapping, Direct Mapping, Set Associative mapping, Virtual memory-Address space and memory Space, Address Mapping using pages, Associative Memory Page Table, I/O Interface, Asynchronous Data Transfer(Strobe Control and handshaking), Modes of transfer, Types of interrupts, Priority Interrupt(Daisy-Chaining Priority, Parallel Priority Interrupt, Priority Encoder, Interrupt Cycle),Direct Memory Access.

Text Books :

1. Computer system Architecture, M.Morris Mano, Pearson Education(3^{rd} Edition).

Reference Books:

1. Computer Organisation, V. Carl Hamacher, Zvonko G Vranesic and Safwat G Zaky, McGraw hill International (4^{th} Edition).
2. Digital Computer Fundamentals, Thomas C Bartee, TMH.

Code:B16 CS 3109

DATABASE MANAGEMENT SYSTEMS
(Elective-I)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To provide an introduction to DBMS, their structure and uses.
2. To prepare SQL commands for creating database objects, populating tables, and retrieve data.
3. To learn ER model
4. To explore the features of RDBMS.
5. To apply conceptual database design, logical database design and normalization.
6. To know different protocols of Concurrency control
7. To explore Recovery techniques of DBMS like ARIES.

Course Outcomes:

After successful completion of the course a student will be able to

1. Generalize the basic concepts of DBMS
2. Explore the relational model
3. Prepare SQL commands for defining, constructing and manipulating databases
4. Apply conceptual and logical database design
5. Apply normalization on tables
6. Schedule concurrent transactions using protocols that provide serializability.
7. Explore techniques for Recovering database

SYLLABUS

Introduction: Introduction to File Processing System and DBMS, Disadvantages of FPS, Advantages of DBMS, Introduction to the Relational model, Integrity constraints, Levels of abstraction, Data Independence, Duties of DBA, Structure of a DBMS.

SQL: SQL Preliminaries, Basic form of SQL Query, Nested Queries, Joins, Set Operations, Aggregate Operators, Integrity Constraints in SQL, Null values, Data Definition Language Commands, Triggers, Views and Sequences, JDBC.

Introduction to Database Design: Basic Steps in Database Design, ER Diagrams, Entities, Attributes and Entity Sets, Relationships & Relationship Sets, Features of the ER Model, Conceptual Database Design and Logical Database Design.

Database Design: Schema Refinement and Normal Forms, Introduction to Schema Refinement, Functional Dependencies, Normal Forms, Properties of Decomposition, Normalization.

Transaction Management : The ACID Properties, Transactions & Schedules, Concurrent Execution of Transactions, Concurrency Control: Lock-Based Concurrency Control, Two Phase Locking protocol, Serializability and Recoverability, Dealing with Deadlocks, Time Stamp Ordering Protocol, Crash Recovery: Log-based Recovery, The Log, Other Recovery-Related Structures, The Write-Ahead Log Protocol, Check pointing, ARIES.

Text Book:

1. Database Management Systems, Raghu Ramakrishnann, Johannes Gehrke, 3rd Edition, McGraw hill, 2014.

Reference Books:

1. Database System Concepts, A. Silberschatz, H.F. Korth, S. Sudarshan, McGraw hill Education, VI edition, 2010.
2. Fundamentals of Database Systems 5th edition. RamezElmasri, Shamkant B. Navathe, Pearson Education, 2008.
3. Database Management System Oracle SQL and PL/SQL, P.K. Das Gupta, PHI.
4. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning, 2008.
5. Database Systems, A Practical approach to Design Implementation and Management Fourth edition, Thomas Connolly, Carolyn Begg, Pearson education.

Code:B16 EE 3106

DIGITAL SIGNAL PROCESSING
(Elective-I)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objective
This course will introduce the basic concepts and techniques for processing signals. By the end of the course, student will be familiar with the most important methods in DSP, including digital filter design, transform-domain processing and importance of Signal Processors.

Course Outcomes
1. Identity properties of discrete-time systems such as time-invariance, stability, causality, and linearity.
2. Compute the linear and circular convolutions of discrete-time sequences.
3. Evaluate and plot the frequency (magnitude and phase) response of linear time-invariant systems.
4. Evaluate the discrete Fourier transform (DFT) of a sequence, relate it to the DTFT,
5. Evaluate the transfer function of linear time-invariant systems & Determine difference equations from transfer function descriptions using Z transforms.
6. Design & Compare IIR & FIR filters

SYLLABUS

Discrete - time signals and systems:
Discrete - time signals – sequences, linear shift – invariant systems, stability and causality, linear constants – coefficient difference equations, frequency domain representation of discrete – time signals and systems.

Applications of z – transforms:
System functions h (z) of digital systems, stability analysis, structure and realization of digital filters, finite word length effects.

Discrete Fourier transform (DFT):
Properties of the dfs, dfs representation of periodic sequences, properties of dft, convolution of sequences.

Fast – Fourier transforms (FFT):
Radix – 2 decimation – in – time (dit) and decimation – in – frequency (dif), fft algorithms, inverse fft.

IIR digital filter design techniques:
Design of iir filters from analog filters, analog filters approximations (butterworth and chebyshev approximations), frequency transformations, general considerations in digital filter design, bilinear transformation method, step and impulse invariance technique.

Design of IIR filters:
Fourier series method, comparison of iir and fir filters.

Text books:

1. Alan v. Oppenheim & Ronald w. Schafer: digital signal processing, phi.

Reference Books:

1. Sanjit k. Mitra, digital signal processing “A computer based approach”, Tata Mc Graw hill.
2. Raddae&rabiner, application of digital signal processing.
3. S. P. Eugene xavier, signals, systems and signal processing, s. Chand & co. Ltd.
4. Antonio, analysis and design of digital filters, tata mc graw hill.

Code: B16 EE 3107

ELECTRICAL MACHINES-I LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Conducting experiments on characteristics of generators & motors
2. Load tests on series,shunt,compound motors and compound generators-swinburne's, Hopkinson's test.
3. OC & SC tests on single phase transformers, Sumpner's test.

Course Outcomes:

1. analyze characteristics of various types of generators & motors which will help in Understanding of machines under various conditions. (K4)
2. compare Speed control of dc motors which will be useful in various industries.(K4)
3. determine testing of machines will give an idea in testing side in various industries(K4).

LIST OF EXPERIMENTS

1. Swimburn's Test
2. Speed control of a DC shunt motor.
3. Load test on DC Shunt motor.
4. Load test on DC series motor.
5. Load test on DC Compound generator.
6. Open circuit characteristics of a DC shunt generator.
7. Hopkinson's Test.
8. Internal and external characteristics of a DC shunt generator.
9. OC and SC tests on a single phase transformer.
10. Load test on a single phase transformer.
11. Sumpner's Test.

Text Books:

1. Kothari.D.P and Nagrath.I.J., "Electrical Machines", Tata McGraw HillPublishingCo.Ltd, New Delhi, 5th edition 2002.
2. Bimbhra.P.S, Electrical Machinery,Khanna Publishers, IL Kosow, "ElectricalMachines& Transformers", Prentice Hall of India.2nd edition 2003.

Reference Books:

1. Electrical Machines, by J B Gupta,S K Kataria& Sons
2. Electrical Machines by U A Bakshi and M V Bakshi, Technical Publications
3. Fitgerald, A.E., Charles Kingsely Jr. Stephen D.Umans, "*Electric Machinery"*McGraw Hill Books Company,6**th**edition 2002**.**
4. Hill Stephen, Chapman.J, "*Electric Machinery Fundamentals"*, McGraw HillBook Co., New Delhi, 4th edition 2005.

Code: B16 EC 3108

LINEAR INTIGRATED CIRCUITS & PULSE DIGITAL CIRCUTS LAB WITH SIMULATION

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

This laboratory course enables students to get practical experience in design, assembly and evaluation of Linear integrated circuits & Pulse Circuits. They will use Multisim to test their electronic designs.

Course Outcomes:
Students will be able to:

1. Design and conduct experiments on RC low pass and high pass circuits.
2. Observe operation of UJT Sweep Generator.
3. Design and test different types of Multi vibrators
4. Acquire a basic knowledge on simple applications of operational amplifier.
5. Design, construct Schmitt trigger using operational amplifier.
6. Use Multisim to test their electronic designs.
7. Design and test different types of Multiplexers and counters.

LIST OF EXPERIMENTS

1. Linear Wave Shaping
   a) Passive RC Differentiator
   b) Passive RC Integrator
2. Non Linear Wave shaping
   a) Clipping Circuits
   b) Clamping Circuits

3) Self bias bistableMultivibrator

4) Schmitt Trigger Using µA 741

5) UJT Sweep Generator

6) AstableMultivibrator using 555 timer

7) Multiplexer

8) Shift Registers

LIST OF EXPERIMENTS
(Simulation)

1 Linear Wave Shaping

- c) Passive RC Differentiator
- d) Passive RC Integrator

2 Non Linear Wave shaping

- c) Clipping Circuits
- d) Clamping Circuits

3) Self bias bistableMultivibrator

4) Schmitt Trigger Using µA 741

5) UJT Sweep Generator

6) AstableMultivibraotr using 555 timer.

7) Multiplexer

8) Shift Registers

Reference: Lab Manuals

Code: B16 ENG3102

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	**: 5 Periods(VA-3+QA-2)**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:
The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA**)**
Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)
Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)
Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms.

Usage: (VA)
Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:
Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:
1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:
1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal.

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency**,** Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1.www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

Code: B16 ENG 3103

BASIC CODING
(Common to ECE & EEE)
(MOOCS-I)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:

At the end of the course students will be able to

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT I Review of Programming constructs
Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers
Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes
Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.
Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT IV User-defined datatypes, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, StructVs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. https://www.geeksforgeeks.org/.
7. https://www.tutorialspoint.com/

Code: B16 EE 3108A

DESIGN OF PHOTOVOLTAIC SYSTEMS
(MOOCS-I)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

WEEK-01 - THE PV CELL

A historical perspective, PV cell characteristics and equivalent circuit, Model of PV cell, Short Circuit, Open Circuit and peak power parameters, Datasheet study, Cell efficiency, Effect of temperature, Temperature effect calculation example, Fill factor, PV cell simulation

WEEK-02 - SERIES AND PARALLEL INTERCONNECTION

Identical cells in series, Load line, Non-identical cells in series, Protecting cells in series, Interconnecting modules in series, Simulation of cells in series, Identical cells in parallel, Non-identical cells in parallel, Protecting cells in parallel, Interconnecting modules in parallel, Simulation of cells in parallel, Measuring I-V characteristics, PV source emulation

WEEK-03 - ENERGY FROM SUN

Insolation and irradiance, Insolation variation with time of day, Earth centric viewpoint and declination, Solar geometry, Insolation on a horizontal flat plate, Energy on a horizontal flat plate, Sunrise and sunset hour angles

WEEK-04 - INCIDENT ENERGY ESTIMATION

Energy on a tilted flat plate, Energy plots in octave, Atmospheric effects, Air Mass, Energy with atmospheric effects, Clearness index, Clearness index and energy scripts in Octave

WEEK-05 - SIZING PV

Sizing PV for applications without batteries, Batteries - Capacity, C-rate, Efficiency, Energy and power densities, Battery selection, Other energy storage methods, PV system design, Load profile, Days of autonomy and recharge, Battery size, PV array size, Design toolbox in octave

WEEK-06 - MAXIMUM POWER POINT TRACKING

MPPT concept, Input impedance of DC-DC converters -Boost converter, Buck converter, Buck-Boost converter, PV module in SPICE , Simulation - PV and DC-DC interface

WEEK-07 - MPPT ALGORITHMS

Impedance control methods, Reference cell, Sampling method, Power slope methods, Hill climbing method, Practical points - Housekeeping power supply, Gate driver, MPPT for non-resistive loads, Simulation

WEEK-08 - PV-BATTERY INTERFACES
Direct PV-battery connection, Charge controller, Battery charger - Understanding current control, slope compensation, simulation of current control, Batteries in series - charge equalisation, Batteries in parallel

WEEK-09 - PELTIER COOLING
Peltier device - principle, Peltier element - datasheet, Peltier cooling, Thermal aspects - Conduction, Convection , A peltier refrigeration example, Radiation and mass transport, Demo of Peltier cooling

WEEK-10 - PV AND WATER PUMPING
Water pumping principle, Hydraulic energy and power, Total dynamic head, Numerical solution - Colebrook formula, Octave script for head calculation, Octave script for hydraulic power, Centrifugal pump, Reciprocating pump, PV power, Pumped hydro application

WEEK-11 - PV-GRID INTERFACE-I
Grid connection principle, PV to grid topologies, 3ph d-q controlled grid connection, dq-axis theory, AC to DC transformations, DC to AC transformations, Complete 3ph grid connection, 1ph d-q controlled grid connection

WEEK-12 - PV-GRID INTERFACE-II and LIFE CYCLE COSTING
SVPWM, Application of integrated magnetics, Life cycle costing, Growth models, Annual payment and present worth factor, LCC with examples

SUGGESTED READING

1. Chenming, H. and White, R.M., Solar Cells from B to Advanced Systems, McGraw Hill Book Co, 1983
2. Ruschenbach, HS, Solar Cell Array Design Hand Varmostrand, Reinhold, NY, 1980
3. Proceedings of IEEE Photovoltaics Specialists Conferences, Solar Energy Journal.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ee35/preview

Code: B16 EE 3108B

ARCHITECTURAL DESIGN OF DIGITAL INTEGRATED CIRCUITS (MOOCS-I)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1: Efficient technique/s for Algorithm to Architecture Mapping

Week 2: Efficient technique/s for Algorithm to Architecture Mapping

Week 3: Recent Trends on Adder/Subtractor Design

Week 4: Recent Trends on Multiplier/Divider Design

Week 5: Efficient VLSI Architectures for Various DSP blocks (FIR filter, CORDIC, FFT etc)

Week 6: Efficient VLSI Architectures for Various DSP blocks (FIR filter, CORDIC, FFT etc)

Week 7: Fundamentals of Efficient Design and Implementation strategies of Digital VLSI Design (Clock Tree synthesis, Timing Closure, Synthesis)

Week 8: Fundamentals of Efficient Design and Implementation strategies of Digital VLSI Design (Clock Tree synthesis, Timing Closure, Synthesis)

SUGGESTED READING MATERIALS:

1. Computer Arithmetic (B. Parhami).
2. Digital Arithmetic (M. D. Ercegovac and T. Lang) .
3. Advanced Arithmetic for the Digital Computer (K. Ulrich)

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ee34/preview

Code: B16 EE 3108C

RECENT ADVANCES IN TRANSMISSION INSULATORS (MOOCS-I)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1: Introduction, Important components of transmission system, Insulation, coordination, Design and selection of insulators for transmission/distribution.

Week 2 : Non ceramic insulators performance-service experience ,Pollution/contamination flashover phenomena, modelling etc ,Failures, importance of reliability and testing.

Week 3 : High Voltage testing and techniques employed ,HV testing techniques for Ceramic/glass Insulators ,HV testing on Composite Insulators.

Week 4 : Surface degradation studies on composite insulators ,Recent studies on composite insulators / Summary.

SUGGESTED READING MATERIALS:

1.Rakosh Das Begamudre, “ Extra High Voltage AC Transmission Engineering”, New Age International(P) Ltd, New Delhi, 2000.
2.E Kuffel, W S Zaengl and J Kuffel, “High Voltage Engg. Fundamentals”, textbook published by Newness publishers, second edition, 2000.
3.Recent IEEE Journal / Conference and CIGRE publications. • “Outdoor Insulators”, Ravi group, Edward Cherney & Jeffery Burnham Text book, 1999
4.Insulators for High Voltage – J S T Looms, text book, Peregrines, 1988 • K Papailiou and F Schmuck, “Silicone Composite Insulators: Materials, design, Applications for Power systems, Springer Berlin Heidelberg, 2012.

WEB LINK:
https://onlinecourses.nptel.ac.in/noc18_ee17/preview

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

ELECTRICAL AND ELECTRONICS ENGINEERING
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EE 3201	Advanced Control Systems	4	3	1	--	4	30	70	100
B16 EE 3202	Power Electronics	4	3	1	--	4	30	70	100
B16 ENG 3201	Principles of Economics & Management	4	3	1	--	4	30	70	100
B16 EE 3203	Power System Protection	4	3	1	--	4	30	70	100
B16 EE 3204	Microprocessor & Microcontroller	4	3	1	--	4	30	70	100
B16 EE 3205	Electrical Machines -II Lab	2	--	--	3	3	50	50	100
B16 EE 3206	Control Systems Lab	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantative Aptitude-II	2	5	--	--	5	100	--	100
B16 EE 3207	Mini Project	2	--	--	3	3	50	--	50
#M-II	MOOCS-II	2	4	--	--	4	100	--	100
#M-III	MOOCS-III	2	4	--	--	4	100	--	100
	Total	**32**	**28**	**5**	**9**	**42**	**600**	**450**	**1050**

#M-II	B16 ENG 3204	Advanced Coding
	B16 EE 3208A	Matlab Programming for Numerical Computation
	B16 EE 3208B	Introduction to Internet of Things
	B16 EE 3208C	Electromagnetic Theory
#M-III	B16 EE 3209A	Industrial Automation And Control
	B16 EE 3209B	Introduction to Smart Grid
	B16 EE 3209C	Analog IC Design
	B16 EE 3209D	Mechanism and RobotKinematics

Code: B16 EE 3201

ADVANCED CONTROL SYSTEMS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. Introducing control system components construction and application.
2. Introducing state variable techniques.
3. To introduce Z-transforms and inverse Z-transform.
4. Introducing classical and modern controllers.
5. To introduce stability tests- Bilinear transformation and Jury's test.

Course Outcomes:

Students will be able to:

1. Know the various components and usage of each component. (K2)
2. Derive stat space model for a given systems and Apply the concept of Observability and Controllability for LTI system. (K3).
3. Apply Z- transform in Engineering application related to digital control systems(K3).
4. Design classical controller based on bode plots and modern controllers based on the state space techniques. (K3)
5. Test the digital system which is useful after designing a particular system with respect to the stability point of view.(K3)

SYLLABUS

CONTROL SYSTEMS COMPONENTS: D.C. & A.C. Tachometers-Synchros, A.C. AndD.C. Servo Motors-Stepper Motors And Its Use In Control Systems, Amplidyne, Metadyne, Magnetic Amplifier –Principle, Operation.

STATE VARIABLE ANALYSIS: Concept Of State, State Variables & State Models, State Model For Linear Continuous Time Systems, Solution Of State Equation, State Transition Matrix, Concept Of Controllability & Observability (Simple Problems To Understand Theory).

THE Z-TRANSFORM: Introduction To Z-Transforms And Inverse Z-Transforms. (Simple Problems To Understand Theory).

INTRODUCTION TO DESIGN: Introduction-Preliminary Considerations OfClassical Design-Lead Compensation-Lag Compensation-Realization Of Compensating Networks-Cascade Compensation In Frequency Domain (Bode Plot Techniques) - Pole Placement By State Feed-Back.

STABILITY: Stability Of Linear Digital Control Systems, Definition & Theorem, Stability Tests, Bi-Linear Transformation Method, Jury’s Stability Test.

Text Books:

1. Control Systems Components By G.J. Gibson &Tuetor
2. Control Systems By R.C. Sukla, Dhanpathrai Publications
3. Control Systems Engineering By I.J. Nagrath&M.Gopal, Wiley Eastern Limited.
4. Automatic Control Systems By B.C. Kuo, Prentice Hall Publication
5. Digital Control Systems By B.C. Kuo, Second Edition, Saunders College Publication-1992

Reference Books:

1. Control System Principles & Design By M. Gopal, Tmh, 1998.
2. Digital Control Systems (Software & Hardware) By Laymount&Azzo
3. Digital Control Systems By Ogata

Code: B16 EE 3202

POWER ELECTRONICS

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To understand the concepts of Power Semiconductor devices and their applications.
2. To understand the need of Energy conversion and effective implementation methods.

Course outcomes:

Student should able to

1. Explain the principle of operation of thyristor, modern power semiconductor devices and necessity of series and parallel connection of thyristors.
2. Explain the operation of Firing and Commutation techniques.
3. Evaluate the phase controlled rectifiers with different loads.
4. Analyze different Choppers, Cycloconverter and AC voltage Controller configurations.
5. Investigate harmonic reduction techniques for inverters based on PWM techniques.

SYLLABUS

Modern Power Semi Conductor Devices

Thyristors – Silicon Controlled Rectifiers (SCRs) – BJT – Power MOSFET – Power IGBT and their characteristics. Basic theory of operation of SCR – Static characteristics and Dynamic characteristics of SCR - Turn on and Turn off times – Turn on and turn off methods. Two transistor analogy of SCR -Series and parallel connections of SCRs Snubber circuit details – Numerical problems.

Thyristorfiring and Commutationcircuits

SCR trigger circuits-R, RC and UJT triggering circuits. The various commutation methods of SCRs-Load commutation- Resonant Pulse Commutation- Complementary Commutation- Impulse Commutation- External Pulse Commutation Techniques. Protection of SCRs

Phase Controlled Rectifiers

Principles of phase controlled rectification -Study of Single phase and three-phase half controlled and full controlled bridge rectifiers with R, RL, RLE loads. Effect of source inductance. Dual converters- circulating current mode and circulating current free mode-control strategies. Numerical problems.

Choppers, Cyclonverter and AC Voltage Controller

Classification of Choppers A, B, C, D and E, Switching mode regulators-Study of Buck, Boost and Buck-Boost regulators, Cukregulators . Principle of operation of Single phase bridge type Cycloconverter and their applications. Single phase AC Voltage Controllers with R and RL loads.

Inverters

Principle of operation of Single phase Inverters -Three phase bridge Inverters (180^0 and 120^0 modes)-voltage control of inverters-Single pulse width modulation- multiple pulse width modulation, sinusoidal pulse width modulation. Harmonic reduction techniues- Comparison of Voltage Source Inverters and Current source Inverters.

Text Books:

1. Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition
2. Power electronics – M.D. Singh & K.B. Kanchandhani, Tata McGraw – Hill Publishing Company, 2nd edition.

Reference Books:

1. Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3rd edition.
2. Power Electronics – VedamSubramanyam, New Age International (p) Limited, Publishers.
3. Power Electronics – P.C. Sen, Tata McGraw-Hill Publishing.
4. Thyristorised power Controllers – G.K. Dubey, S.R Doradra, A. Joshi and R.M.K. Sinha, New Age international Pvt Ltd. Publishers latest edition

Code: B16 ENG 3201

PRINCIPLES OF ECONOMICS AND MANAGEMENT

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course objectives

1. To familiarise Engineering students with the real world of business.
2. To provide capacity to analyse business decisions by understanding the origin & concepts of economics.
3. To provide capacity to analyse business decisions by understanding the concepts of management.
4. To help engineering students know the need and importance of entrepreneurship.

Course outcomes

1. Students will be able to gain empirical knowledge and understand the complete frame work of business.
2. To analyse the concepts pertaining to economic decision making.
3. To analyse the concepts of Managerial decision making.
4. To inculcate the spirit of Entrepreneurship and gain knowledge for setting up an enterprise.

SYLLABUS

Introduction to Managerial Economics:
Wealth, Welfare and Scarce Definitions of Economics, Micro and Macro Economics, Demand –Law of Demand, Elasticity of Demand, Types of Elasticity and Factors Determining price Elasticity; Demand :Utility-Law of Diminishing Marginal Utility and its limitations.

Conditions of Different Market Structures:
Perfect Competition, Monopolistic Competition, Monopoly, Oligopoly and Duopoly.

Forms of Business Organisation:
Sole Proprietorship, Partnership, Joint Stock Company-Private Limited and public limited Companies. Public Enterprise and their types.

Introduction to Management:
Functions of Management – Taylor’s Scientific Management; Henry Fayol’s Principles of Management;

Human Resource Management:
Basic functions of HR Manager; Man Power Planning, Recruitment, Selection, Training, Development, Placement, Compensation and Performance Appraisal (In Brief).

Production Management:
Production Planning and Control, Plant Location, Break-Even Analysis, Assumptions and Applications.

Financial Management:
Types of capital; Fixed and Working Capital and Methods of Raising Finance; Depreciation; Straight Line and Diminishing Balance Methods.

Marketing Management:
Functions of Marketing and Distribution channels

Entrepreneurship-
Entrepreneurial Functions, Entrepreneurial Development; Objectives, Training, Benefits; Phase of Installing a Project.

Text Books:

1. K KDewtt, Modern Economic Theory ,S.Chand& Company New Delhi-55
2. S.C.Sharma and Banga T.R, Industrial Organisation and engineering EconomicsKhanna Publications Delhi-6

Reference Books:

1. A.R.Araryasri Management Science,Tata McGraw-Hill,New Delhi-20
2. A.R.Araryasri, Managerial Economics and financial analysis, Tata McGraw-Hill,New Delhi-20

Code: B16 EE 3203

POWER SYSTEM PROTECTION

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To know the need for protection, various devices for protection and to be familiar with the terminology used in the power system protection.
2. To understand the working of various types of fuses and circuit breakers and various challenges during their operation.
3. To know the principle of operation,classification, constructional features and testing of circuit breakers.
4. To be familiar with the principle of operation and different types of protective relays and their application to alternators, transformers and feeders.
5. To understand the concepts of over voltage protection and devices used for overvoltage protection.

Course Outcomes:

Students are able to

1. Identify the need for protection and know various devices for protection and terminology used in protection.
2. Discriminate the constructional details with operation principle of various types of fuses, circuit breakers, relays, lightning arresters and their applications.
3. Apply the arc quenching methods to various types of circuit breakers.
4. Apply various relays to various types of power system equipment like alternator, transformer and feeders and distinguish between an electromagnetic relay and a static relay.
5. Identify the different causes for over voltages and choose various protection devices against over voltages.

SYLLABUS

Introduction to Power System Protection: Need for protective systems, Nature and causes of faults, Types and effects of faults, Fault statistics, Evolution of protective relays, Zones of protection, Primary and Back-up protection, Essential qualities of protection, Classification of protective relays, Classification of Protective Scheme, CTs and PTs and their applications in protection schemes, Summation transformer, Phase-sequence current segregating network, Basic relay terminology.

Fuses and Circuit Breakers: Fuses and their types, High-voltage HRC fuses and their applications, Selection of fuses. Circuit breakers, Formation of arc, Methods of arc extinction, Restriking voltage, Recovery voltage, RRRV, Single and double frequency transients, Resistance switching, Current chopping, Switching of capacitor banks and un-loaded lines, Ratings and characteristics of circuit breakers.

Types of Circuit Breakers and Testing: Principle of operation of circuit breakers, Classification of circuit breakers, Constructional Features of Air Circuit Breakers, Oil Circuit Breakers, Air Blast Circuit Breakers, SF-6 Circuit Breakers and Vacuum Circuit Breakers, Testing of Circuit Breakers.

Protective Relays: Different types of protective relays, Principle of operation and characteristics of relays, Overcurrent, Earth fault and Phase fault protection, Differential and Distance protection with simple applications to Alternators; Transformers; Single and parallel feeders. Introduction to Static relaying, Static relays for time lag Overcurrent and Differential Protection.

Overvoltage Protection: Causes of over voltages, Over voltages due to Lightning, Protection against Lightning and Travelling Waves – Earth Wire, Spark Gap, Surge Arresters, Lightning Arresters, Surge Absorber, Peterson Coil, Insulation Co-ordination.

Text Books:

1. "Power System Protection and Switchgear" by Badri Ram and D. N. Vishwakarma, Tata McGraw-Hill Education, 2001.

Reference Books:

1. "Electrical Power Systems" by C. L. Wadhwa, New Age International, 2009.
2. "Power System Protection" by Paul M Anderson, Wiely, 1998.
3. "Switchgear Protection and Power Systems", by Sunil S. Rao, Khanna Publishers, 2008.

Code: B16 EE 3204

MICROPROCESSOR & MICROCONTROLLER

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course objectives

1. Study the architecture of 8085 microprocessor, functions, instruction sets etc.
2. Architecture of different peripherals like 8259,8255,8279 and A/D and D/A convertors
3. Study the architecture of 8051 microcontroller, functions etc.
4. study 8086 architecture and addressing.

Course Outcomes

1. Understand thefundamentals of 8085 Microprocessor and microcontroller based systems.
2. Familiarize with the instruction set and assembly level programming.
3. Illustrate how the different peripherals (8255, 8253 etc.) are interfaced with Microprocessor.
4. Distinguish and analyze the properties of Microprocessors & Microcontrollers.
5. Apply knowledge on interfacing microcontrollers for some real time applications.

SYLLABUS

8085 Microprocessor:
Introduction to microprocessors, micro computers – Architecture of 8085 microprocessor – pin-out diagram of 8085 – Detailed description of the 8085 pins – addressing modes Memory interfacing – Machine cycles and bus timings for Opcodefetch,memory read, memory write, I/O read, I/O write operations – Memory mapped I/O and I/O mapped I/O.

8085 Instructions and programming:
Difference between Machine language, Assembly language and High level language – Brief description of the 8085 instruction set – 8085 programming using data transfer group, arithmetic group, logical group, branch transfer group, stack and subroutines – counters and delay .

Interfacing peripherals to 8085:
Function of D/A and A/D converters – Interfacing D/A and A/D converters. Detailed description and interfacing of 8251 USART, 8253/8254 programmable timer, 8255 PPI, 8257 DMA controller, 8279 programmable keyboard/display interface

8051 Microcontroller:
Introduction to microcontrollers – Comparison between microprocessors and microcontrollers – Functional block diagram of 8051 microcontroller and its description – 8051 pin-out diagram and description of 8051 pins – Interfacing external memory to 8051 – implementing counters and timers in 8051 – Serial data transfer using 8051 – Various interrupts and its programming in 8051. Interfacing Stepper motor, to 8051 microcontroller.

Advanced topics in Microprocessors:
Architecture of 8086 microprocessor pin out diagram – Addressing modes – differences between 8085 and 8086.

Text Books:

1. Ramesh S. Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publishing, New Delhi, 5th edition, 2008.
2. Kenneth. J.Ayala, The 8051 Microcontroller, Cengage learning, 3rd edition ,2005.
3. A.K.Ray and Bhurchandi, "Advanced Microprocessors and Peripherals", 2nd Edition, TMH Publications

Reference Books:

1. Douglas V.Hall, "Microprocessors and Interfacing", 2nd Revised Edition, TMH Publications.Design", 2nd ed., PHI
2. Krishna Kant, "Microprocessors and Microcontrollers", PHI Publisher.

Code: B16 EE 3205

ELECTRICAL MACHINES-II LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Students conduct experiments on characteristics of synchronous and induction machines.

Course Outcomes:

Students are able to

1. Calculate the regulation of an alternator by EMF,MMF and ZPF methods [K3]
2. Verify Alternator synchronism and draw the performance characteristics, finding out different reactances. [K4]
3. Find the efficiency and machine performances by conducting various tests on 3-Φ and 1-Φ induction motor. [K3]
4. Verify the speed variation of induction machine. [K4]

List of Experiments

1. NL and BR test on a three phase squirrel cage induction motor.
2. Regulation of alternator by EMF and MMF methods.
3. Regulation of alternator by ZPF method.
4. Characterisitcs of line excited induction generator.
5. Characteristics of induction start synchronous motor.
6. Load test on three phase slip ring induction motor.
7. V and inverted V curves of synchronous motor.
8. Measurement of X_d and X_q of a synchronous machine.
9. Equivalent circuit of a single phase induction motor.
10. Measurement of sequence reactance of a synchronous machine.

Text Books:

1. Nagrath& Kothari," Electric Machines" TMH
2. PS Bhimbra, "Electrical Machinery", Khanna Publishers.

Reference Books:

1. Fitzgerald & Kingsley, "Electric Machinery" McGraw Hill
2. Alexander S. Langsdorf, "AC Machines", Tata McGraw Hill.
3. MG Say, "Theory Performance and Design of AC Machines" CBS Publisher

Code: B16 EE 3206

CONTROL SYSTEMS LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course objectives:

1. They get the basic knowledge on practical control system.
2. They get the knowledge on applications of machines & electronic devices with control systems.

Course outcomes:

1. Will be able to do various engineering projects.
2. Ability to formulate transfer function for given control system problems.
3. Ability to find time response of given control system model.
4. Plot Root Locus and Bode plots for given control system model
5. Ability to design Lead, Lag, Lead-Lag systems in control systems
6. Ability to design PID controllers for given control system model

List of experiments:

1. Magnetic amplifier
2. Study of DC Servo motor
3. DC Position control system
4. Study of first order system
5. Study of second order system
6. Speed torque characteristics of AC Servomotor
7. PID Controller
8. Synchro Transmitter and Receive pair
9. Study of digital control system
10. Study of Lead-Lag compensators

Reference books:

1. Control Systems Engineering - I. J. Nagrath and M. Gopal, New Age International (P) Limited, Publishers.
2. Control Systems - A. Ananad Kumar, PHI
3. Control Systems Theory and Applications - S. K. Bhattacharya, Pearson.
4. Control Systems - N. K. Sinha, New Age International (P) Limited Publishers.

Code: B16ENG3202

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	**: 5 Periods(VA-3+QA-2)**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)
Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)
Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)
Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)
Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)
Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:
1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods, Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems-circular arrangement, linear arrangement, different puzzles.

UNIT V:Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Code: B16 EE 3207

MINI PROJECT

Lab: 3 Hrs. **Sessionals : 50**
Credits : 2

Course objectives:

1. To make active participation of the student in identifying and finding the solutions involved in the making of project.
2. Gain knowledge on various softwares,micro controllers and different components.
3. To improve problem solving skills and managing skills of the student.

Course out comes:

The student will be able to

1. Identify the area of project work through the literature survey.
2. Plan the project activity with constraints required to implement it.
3. Develop communication and presentation skills.
4. Work as team member for core/multidisciplinary projects.

Code: B16 ENG 3204

ADVANCED CODING
(Common to ECE & EEE)
(MOOCS-III)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS

UNIT I Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. https://www.geeksforgeeks.org/.
7. https://www.tutorialspoint.com/

Code: B16 EE 3208A

MATLAB PROGRAMMING FOR NUMERICAL COMPUTATION (MOOCS-II)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Module 1: Introduction to MATLAB Programming

This module will introduce the students to MATLAB programming through a few examples. Students who have used MATLAB are still recommended to do this module, as it introduces MATLAB in context of how we use it in this course

Lecture 1-1 Basics of MATLAB programming
Lecture 1-2 Array operations in MATLAB
Lecture 1-3 Loops and execution control
Lecture 1-4 Working with files: Scripts and Functions
Lecture 1-5 Plotting and program output

Module 2: Approximations and Errors

Taylor's / Maclaurin series expansion of some functions will be used to introduce approximations and errors in computational methods
Lecture 2-1 Defining errors and precision in numerical methods
Lecture 2-2 Truncation and round-off errors
Lecture 2-3 Error propagation, Global and local truncation errors

Module 3: Numerical Differentiation and Integration

Methods of numerical differentiation and integration, trade-off between truncation and round-off errors, error propagation and MATLAB functions for integration will be discussed.
Lecture 3-1 Numerical Differentiation in single variable
Lecture 3-2 Numerical differentiation: Higher derivatives
Lecture 3-3 Differentiation in multiple variables
Lecture 3-4 Newton-Cotes integration formulae
Lecture 3-5 Multi-step application of Trapezoidal rule
Lecture 3-6 MATLAB functions for integration

Module 4: Linear Equations

The focus of this module is to do a quick introduction of most popular numerical methods in linear algebra, and use of MATLAB to solve practical problems.
Lecture 4-1 Linear algebra in MATLAB
Lecture 4-2 Gauss Elimination
Lecture 4-3 LU decomposition and partial pivoting
Lecture 4-4 Iterative methods: Gauss Siedel
Lecture 4-5 Special Matrices: Tri-diagonal matrix algorithm

Module 5: Nonlinear Equations
After introduction to bisection rule, this module primarily covers Newton-Raphson method and MATLAB routines fzero and fsolve.
Lecture 5-1 Nonlinear equations in single variable
Lecture 5-2 MATLAB function fzero in single variable
Lecture 5-3 Fixed-point iteration in single variable
Lecture 5-4 Newton-Raphson in single variable
Lecture 5-5 MATLAB function fsolve in single and multiple variables
Lecture 5-6 Newton-Raphson in multiple variables

Module 6: Regression and Interpolation
The focus will be practical ways of using linear and nonlinear regression and interpolation functions in MATLAB.
Lecture 6-1 Introduction
Lecture 6-2 Linear least squares regression(including lsqcurvefit function)
Lecture 6-3 Functional and nonlinear regression (including lsqnonlin function)
Lecture 6-4 Interpolation in MATLAB using spline and pchip

Module 7: Ordinary Differential Equations (ODE) – Part 1
Explicit ODE solving techniques in single variable will be covered in this module.
Lecture 7-1 Introduction to ODEs; Implicit and explicit Euler's methods
Lecture 7-2 Second-Order Runge-Kutta Methods
Lecture 7-3 MATLAB ode45 algorithm in single variable
Lecture 7-4 Higher order Runge-Kutta methods
Lecture 7-5 Error analysis of Runge-Kutta method

Module 8: Ordinary Differential Equations (ODE) – Practical aspects
This module will cover ODE solving in multiple variables, stiff systems, and practical problems. The importance of ODEs in engineering is reflected by the fact that two modules are dedicated to ODEs.
Lecture 8-1 MATLAB ode45 algorithm in multiple variables
Lecture 8-2 Stiff ODEs and MATLAB ode15s algorithm
Lecture 8-3 Practical example for ODE-IVP
Lecture 8-4 Solving transient PDE using Method of Lines

REFERENCE MATERIALS
Textbook:
Fausett L.V. (2007) Applied Numerical Analysis Using MATLAB, 2nd Ed., Pearson Education
Reference Book:
Chapra S.C. and Canale R.P. (2006) Numerical Methods for Engineers, 5th Ed., McGraw Hill

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ge05/course

Code: B16 EE 3208B

INTRODUCTION TO INTERNET OF THINGS
(MOOCS-II)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1: Introduction to IoT: Part I, Part II, Sensing, Actuation, Basics of Networking: Part-I

Week 2: Basics of Networking: Part-II, Part III, Part IV, Communication Protocols: Part I, Part II

Week 3: Communication Protocols: Part III, Part IV, Part V, Sensor Networks: Part I, Part II

Week 4: Sensor Networks: Part III, Part IV, Part V, Part VI, Machine-to-Machine Communications

Week 5: Interoperability in IoT, Introduction to Arduino Programming: Part I, Part II, Integration of Sensors and Actuators with Arduino: Part I, Part II

Week 6: Introduction to Python programming: Part I, Part II, Introduction to Raspberry Pi: Part I, Part II, Implementation of IoT with Raspberry Pi: Part I

Week 7: Implementation of IoT with Raspberry Pi: Part II, Part III, Introduction to SDN: Part I, Part II, SDN for IoT: Part I

Week 8: SDN for IoT: Part II, Data Handling and Analytics: Part I, Part II, Cloud Computing: Part I, Part II

Week 9: Cloud Computing: Part III, Part IV, Part V, Sensor-Cloud: Part I, Part II

Week 10: Fog Computing: Part I, Part II, Smart Cities and Smart Homes: Part I, Part II, Part III

Week 11: Connected Vehicles: Part I, Part II, Smart Grid: Part 1, Part II, Industrial IoT: Part I

Week 12: Industrial IoT: Part I, Case Study: Agriculture, Healthcare, Activity Monitoring: Part I, Part II

SUGGESTED READING

1. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press)
2. "Internet of Things: A Hands-on Approach", by ArshdeepBahga and Vijay Madisetti (Universities Press)
3. Research papers

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs46/course

Code: B16 EE 3208C

ELECTROMAGNETIC THEORY
(MOOCS-II)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1 : Coulomb's law and electric fields

Week 2 : Gauss's law, potential and energy, conductors and dielectrics

Week 3 : Laplace and Poisson equations, solution methods, and capacitance

Week 4 : Biot-Savart and Ampere's laws, inductance calculation

Week 5 : Magnetic materials, Faraday's law and quasi-static analysis

Week 6 : Maxwell equations and uniform plane waves

Week 7 : Wave propagation in dielectrics and conductors, skin effect, normal incidence

Week 8 : Oblique incidence, Snell's law, and total internal reflection

Week 9 : Transmission lines, Smith chart, impedance matching

Week 10 : Transients and pulse propagation on transmission line

Week 11 : Waveguides: Metallic and Dielectric

Week 12 : Antenna fundamentals

SUGGESTED READING MATERIALS:

- Engineering Electromagnetics, W. H. Hayt and J. A. Buck, 7th edition, Tata McGraw Hill
- Electromagnetics with applications, J. D. Kraus and Fleisch, Tata McGraw-Hill
- Principles of Electromagnetics, M. O. Sadiku, Oxford University Press
- Fields and Waves in Communication Electronics, S. Ramo, J. R. Whinnery, and T. Van Duzer, Wiley

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ee04/course

Code: B16 EE 3209A

INDUSTRIAL AUTOMATION AND CONTROL (MOOCS-III)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Module I
1 Introduction
2 Introduction(Cont.)
3 Architecture of Industrial Automation Systems
4 Architecture of Industrial Automation Systems(Cont.)
Module II
5 Measurement Systems Characteristics
6 Measurement Systems Characteristics(Cont.)
7 Data Acquisition Systems
8 Data Acquisition Systems (Cont.)
Module III
9 Introduction to Automatic Control
10 Introduction to Automatic Control(Cont.)
11 P-I-D Control
12 P-I-D Control(Cont.)
13 PID Control Tuning
14 PID Control Tuning(Cont.)
15 Feedforward Control Ratio Control
16 Feedforward Control Ratio Control(Cont.)
17 Time Delay Systems and Inverse Response Systems
18 Time Delay Systems and Inverse Response Systems(Cont.)
19 Special Control Structures
20 Special Control Structures(Cont.)
21 Concluding Lesson on Process Control (Self-study)
22 Introduction to Sequence Control, PLC , RLL
23 Introduction to Sequence Control, PLC, RLL(Cont.)
24 Sequence Control. Scan Cycle, Simple RLL Programs
25 Sequence Control. Scan Cycle, Simple RLL Programs(Cont.)
26 Sequence Control. More RLL Elements, RLL Syntax
27 Sequence Control. More RLL Elements, RLL Syntax(Cont.)
28 A Structured Design Approach to Sequence Control
29 A Structured Design Approach to Sequence Control(Cont.)
30 PLC Hardware Environment
31 PLC Hardware Environment(Cont.)
Module IV
32 Flow Control Valves
33 Flow Control Valves(Cont.)
34 Hydraulic Control Systems - I
35 Hydraulic Control Systems – I(Cont.)
36 Hydraulic Control Systems - II
37 Hydraulic Control Systems – II(Cont.)
38 Industrial Hydraulic Circuit
39 Industrial Hydraulic Circuit(Cont.)
40 Pneumatic Control Systems - I

41 Pneumatic Control Systems – I(Cont.)
42 Pneumatic Systems - II
43 Pneumatic Systems – II(Cont.)
44 Energy Savings with Variable Speed Drives
45 Energy Savings with Variable Speed Drives(Cont.)
46 Introduction To CNC Machines
47 Introduction To CNC Machines(Cont.)

Module V

48 The Fieldbus Network - I
49 The Fieldbus Network - I(Cont.)
50 Higher Level Automation Systems
51 Higher Level Automation Systems(Cont.)
52 Course Review and Conclusion (Self-study)

SUGGESTED READING

1.Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and A. K. Deb, Jaico Publishing House, 2013.

2. Chemical Process Control, an Introduction to Theory and Practice, George Stephanopoulos, Prentice Hall India, 2012.

3.Electric Motor Drives, Modelling, Analysis and Control, R. Krishnan, Prentice Hall India, 2002.

4.Hydraulic Control Systems, Herbert E. Merritt, Wiley, 1991

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ee12/course

Code: B16 EE 3209B

INTRODUCTION TO SMART GRID
(MOOCS-III)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1 :

- Introduction to Smart Grid-I
- Introduction to Smart Grid-II
- Architecture of Smart Grid System
- Standards for Smart Grid System
- Elements and Technologies of Smart Grid System

Week 2 :

- Elements and Technologies of Smart Grid System-II
- Distributed Generation Resources-I
- Distributed Generation Resources-II
- Distributed Generation Resources-III
- Distributed Generation Resources-IV

Week 3 :

- Wide Area Monitoring Systems-I
- Wide Area Monitoring Systems-II
- Phasor Estimation-I
- Phasor Estimation-II
- Digital relays for Smart Grid Protection

Week 4 :

- Islanding Detection Techniques-I
- lslanding Detection Techniques-II
- Islanding Detection Techniques-III
- Islanding Detection Techniques-IV
- Smart Grid Protection-I

Week 5 :

- Smart Grid Protection-II
- Smart Grid Protection-III
- Modelling of Storage Devices
- Modelling of DC Smart Grid components
- Operation and control of AC Microgrid-I

Week 6 :

- Operation and control of AC Microgrid-II
- Operation and control of DC Microgrid-I
- Operation and control of DC Microgrid-II
- Operation and control of AC-DC hybrid Microgrid-I
- Operation and control of AC-DC hybrid Microgrid-II

Week 7 :

- Simulation and Case study of AC Microgrid
- Simulation and Case study of DC Microgrid
- Simulation and Case Study of AC-DC Hybrid Microgrid
- Demand side management. of Smart Grid
- Demand response analysis of Smart Grid

Week 8 :

- Energy Management
- Design of Smart grid and Practical Smart Grid case study-I
- Design of Smart grid and Practical Smart Grid case study-II
- System Analysis of AC/DC Smart Grid
- Conclusions

SUGGESTED READING MATERIALS:

1.Smart power grids by A Keyhani, M Marwali.
2.Computer Relaying for Power Systems by ArunPhadke
3.Microgrids Architecture and control by Nikos Hatziargyriou
4.Renewable Energy Systems by Fang Lin Luo, Hong Ye
5.Voltage-sourced converters in power systems_ modeling, control, and applications by AmirnaserYazdani, Reza Iravani"

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ee42/preview

Code: B16 EE 3209C

ANALOG IC DESIGN
(MOOCS-III)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1 Introduction to Integrated Circuits; Review of some simple MOS amplifiers

Week 2 Noise in circuits; Noise analysis of common circuits

Week 3 Mismatch in circuits

Week 4 Introduction to negative feedback; frequency response and Bode plots

Week 5 Loop gain and stability; Dominant pole compensation

Week 6 Block level conceptualization of single- and two-stage opamps

Week 7 Differential and common-mode analysis; Differential amplifiers

Week 8 Single-stage opamp

Week 9 Telescopic opamp

Week 10 Folded cascodeopamp

Week 11 Two-stage opamp

Week 12 Fully differential opamps; common-mode feedback

SUGGESTED READING

- Design of Analog CMOS Integrated Circuits by BehzadRazavi; Tata McGraw-Hill Edition 2006 (ISBN: 0070529035).

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ec05/course

Code: B16 EE 3209D

MECHANISM AND ROBOT KINEMATICS
(MOOCS-III)

Theory	**: 4 Periods**	**Sessionals**	**: 100**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Week 1: Introduction to Mechanisms and Robotics, Mobility Analysis-I

Week 2: Mobility Analysis-II, Displacement Analysis: constrained mechanisms and robots-I

Week 3: Displacement Analysis: constrained mechanisms and robots-II

Week 4: Displacement Analysis: constrained mechanisms and robots- III, Velocity Analysis: constrained mechanisms and robots-I

Week 5: Velocity Analysis: constrained mechanisms and robots-II

Week 6: Velocity Analysis: constrained mechanisms and robots-III

Week 7: Velocity Analysis: singularity and path generation, Acceleration Analysis, Force Analysis-I

Week 8: Force Analysis-II, Coordinate Transformations and kinematics of serial robots

REFERENCES:

1. Theory of Mechanisms and Machines: A. Ghosh and A.K. Mallik
2. Kinematics, Dynamics and Design of Machinery: K.J. Waldron and G.L. Kinzel
3. Robotics: K.S. Fu, R.C. Gonzalez and C.S.G. Lee
4. Foundations of Robotics: T. Yoshikawa

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me18/course

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
Recognised as Scientific and Industrial Research Organisation
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ESTD: 1980

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

ELECTRICAL AND ELECTRONICS ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Sessional Marks	Exam Marks	Total Marks
B16 EE 4101	Electric Drives	4	3	1	-	4	30	70	100
#ELE-II	Elective-II	4	3	1	-	4	30	70	100
#ELE-III	Elective-III	4	3	1	-	4	30	70	100
B16 EE 4108	Microprocessor & Microcontroller Lab	2	-	-	3	3	50	50	100
B16 EE 4109	Power Electronics Lab	2	-	-	3	3	50	50	100
B16 EE 4110	Project Phase-I	2	--	--	3	3	50	--	50
	Total	**18**	**9**	**3**	**9**	**21**	**240**	**310**	**550**

Elective-II	B16 EE 4102	Non-Conventional Energy Sources
	B16 EE 4103	Energy Management and Auditing
	B16 EE 4104	Electrical Machine Design
Elective-III	B16 EE 4105	Operations Research
	B16 EE 4106	Flexible AC Transmission Systems
	B16 EE 4107	Introduction to Soft Computing

Code:B16EE4101

ELECTRIC DRIVES

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To understand the concept of drive and multi-quadrant operation of drive.
2. It covers in detail the basic and advanced speed control techniques using power electronic converters that are used in industry.
3. To understand the operation of Rectifier and Chopper fed DC drives.
4. Describes the slip power recovery schemes in induction motors and operation of AC drives.

Course Outcomes:

Upon the completion of this course, the student will be able to

1. Identify different electric drive system.
2. Understand the operation of rectifier fed DC drives, chopper fed DC drives and closed loop control of DC motor.
3. Analyse the slip power recovery schemes of Induction motor and speed control of converter fed induction motor & synchronous motor.
4. Evalute the performance of speed control of synchronuous motor by CSI and VSI.

SYLLABUS

INTRODUCTION TO DRIVES Definition, Advantages and applications of drives, Components of electric drive system, Difference between DC and AC drives, Multi quadrant operation of drive, Speed control methods of DC motors and Induction motor, Starting methods of synchronous motor, Electric Braking.

RECTIFIER CONTROLLED FED DC DRIVES Single Phase Fully controlled converters connected to DC separately excited motor and DC series motor – Continuous & Discontinuous current operation – voltage and current waveforms – Speed Torque expressions – Speed Torque Characteristics.

CHOPPER CONTROLLED FED DC DRIVES Chopper controlled DC separately excited motor and DC series motor – Continuous current operation – voltage and current waveforms – Speed Torque expressions – Speed Torque characteristics, Closed loop control of DC drive (Only Block Diagram).

CONTROL OF INDUCTION MOTORS Variable voltage control of Induction motor by AC voltage controller, Variable frequency control of Induction motor by cycloconverter – waveforms – Speed Torque characteristics, Slip power recovery schemes – Static Kramer Drive – Static Scherbius Drive.

CONTROL OF SYNCHRONOUS MOTORS Separate control & self-control of synchronous motors – Operation of self-controlled synchronous motors by VSI and CSI cycloconverters, Load commutated CSI fed Synchronous Motor – Operation – Waveforms – Speed Torque characteristics.

Text Books:

1. Fundamentals of Electrical Drives by G.K.Dubey, Second Edition, 2002.
2. Power Electronics: Circuits, Devices and Applications by M.H.Rashid, Third Edition, 2009.

Reference Books:

1. Power Electronics by M.D.Singh and K.B.Khanchandani, Second Edition, 2017.
2. Modern Power Electronics and AC Drives by Bimal K Bose, 2005.
3. Thyristor Control of Electric Drives by Vedam Subramanyam, Tata McGraw-Hill Publications, 2008.

Code:B16EE4102

NON-CONVENTIONAL ENERGY SOURCES
(Elective-II)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. To know the need for Renewable energy.
2. To understand the concepts of solar energy and its conversion systems.
3. To Understand the basic physics of wind power generation.
4. To be familiar with the ocean, wave and tide energies.
5. To understand the concepts of geo-thermal energy and bio energy.

Course Outcomes:

After doing this course, the students will be able to

1. Identify the need for Renewable energy
2. Recognise the ways of collection of solar energy.
3. Apply the knowledge of wind energy to estimate the energy potential.
4. Apply the knowledge of ocean, waves and tides to estimate their energy potential.
5. Understand the concepts behind geo-thermal energy and bio energy.

Introduction to Non-Conventional Energy Sources:

Environmental aspects of conventional electric energy generation, renewable and non-conventional energy sources, impact of renewable energy generation on environment, prospects of renewable energy sources.

Solar Energy:

Solar radiation and its measurements: introduction to solar energy, solar constant, solar radiation at the earth's surface, solar radiation geometry, solar radiation measurements, estimation of average solar radiation, solar radiation on tilted surface. Solar energy collectors: physical principles of the conversion of solar radiation into heat, flat plate collectors, concentrating collectors, advantages and disadvantages. Solar electric power generation: principles of solar photo-voltaic cells, conversion efficiency and power output.

Wind Energy:

Introduction, basic principles of wind energy conversion-nature of wind, power in the wind, maximum power, forces on the blades, lift and drag forces, aerodynamics, types of wind power plants, types of wind turbine - generating units, generating systems, energy storage, application of wind energy, site selection considerations, environmental aspects.

Ocean Energy: Ocean thermal energy conversion: working principle, availability, types, advantages, limitations and applications. Wave energy: factors affecting the wave energy, mathematical analysis for potential energy, kinetic energy, total energy and wave power. Tidal energy: basic terminology, types of tidal plants, energy potential estimation from a tidal plant, advantages and limitations.

Geo-Thermal Energy: Structure of earth's interior, thermal gradient, geo-thermal energy sources, types of geo-thermal power generation, merits, demerits and applications of geo-thermal energy.

Bio Energy**:** Overview, bio-mass conversion processes, bio-gas generation, factors affecting the generation of bio gas, various types of bio gas plants.

Text Books:

1. G. D. Rai, "Non-Conventional Energy Sources", fifth edition, Khanna Publishers, 2015.
2. D. P. Kothari, K. C. Singal and Rakesh Ranjan, "Renewable Energy sources and Emerging Technologies", 2^{nd} Edition, PHI Learning Pvt. Limited, 2013.
3. S. Hasan Saeed and D. K. Sharma, "Non-Conventional Energy Resources", First Edition, S.K. Kataria & Sons, 2013.

Reference Books:

1. S. P. Sukhatme, "Solar Energy", Third edition, Tata McGraw-Hill Education, 1996.
2. S. Sumathi, L. Ashok Kumar and P. Surekha, "Solar PV and Wind Energy Conversion Systems (Green Energy and Technology)", First Edition, Springer, 2015
3. G. N. Tiwari and M. K. Ghosal, "Renewable energy resources", First Edition, Narosa Publishing House, 2004.

Code:B16EE4103

ENERGY MANAGEMENT AND AUDITING
(Elective-II)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students are able to

1. Understand importance of energy and energy security.
2. Understand impact of use energy resources on environment and emission standards, different operating frame work.
3. Follow format of energy management, energy policy.
4. Learn various tools of Demand Control.
5. Calculate economic viability of energy saving option

Course Outcomes:

1. Analyze and understand energy consumption patterns and environmental impacts and mitigation method.
2. Listing various energy conservation measures for various processes.
3. Students can carry out preliminary audits.
4. Can work out economic feasibility of encon option.

SYLLABUS

ENERGY SCENARIO

Classification of Energy resources, Commercial and non-commercial energy, primary and secondary sources, commercial energy production, final energy consumption, Energy needs of growing economy, short terms and long terms policies, energy sector reforms, distribution system reforms and up-gradation, energy security, importance of energy conservation, energy and environmental impacts, emission check standard, United nations frame work convention on climate change, Global Climate Change Treaty, Kyoto Protocol, Clean Development Mechanism, salient features of Energy Conservation Act 2001 and Electricity Act 2003. Indian and Global energy scenario. Introduction to IE Rules. Study of Energy Conservation Building Code (ECBC), Concept of Green Building.

ENERGY MANAGEMENT

Definition and Objective of Energy Management, Principles of Energy management, Energy management Strategy, Energy Manager Skills, key elements in energy management, force field analysis, energy policy, format and statement of energy policy, Organization setup and energy management. Responsibilities and duties of energy manager under act 2001. Energy Efficiency Programmes. Energy monitoring systems. Introduction to SCADA and Automatic meter reading in utility energy management.

DEMAND MANAGEMENT

Supply side management (SSM), various measures involved such as use of FACTS, VAR Compensation, Generation system up gradation, constraints on SSM. Demand side management (DSM), advantages and Barriers, implementation of DSM, areas of development of demand side management in agricultural, domestic and commercial consumers. Demand management through tariffs (TOD). Power factor penalties and incentives in tariff for demand control. Apparent energy tariffs. Role of renewable energy sources in energy management, direct use (solar thermal, solar air conditioning, biomass) and indirect use (solar, wind etc.)

ENERGY AUDIT

Definition, need of energy audit, types of audit, procedures to follow, data and information analysis, energy audit instrumentation, energy consumption – production relationship, pie charts. Sankey diagram, Cusum technique, least square method and numerical based on it. Outcome of energy audit and energy saving potential, action plans for implementation of energy conservation options. Bench- marking energy performance of an industry. Energy Audit Report writing as per prescribed format. Audit case studies of sugar, steel, paper and cement industries.

FINANCIAL ANALYSIS AND CASE STUDIES

Costing techniques; cost factors, budgeting, standard costing, sources of capital, cash flow diagrams and activity chart. Financial appraisals; criteria, simple payback period, return on investment, net present value method, time value of money, break even analysis, sensitivity analysis and numerical based on it, cost optimization, cost of energy, cost of generation, Energy audit case studies such as IT sector, Textile, Municipal corporations, Educational Institutes, T and D Sector and Thermal Power stations.

Text Books:

1. Guide books for National Certification Examination for Energy Managers/Energy Auditors Book 1, General Aspects (available on online)
2. Guide books for National Certification Examination for Energy Managers/Energy Auditors Book 2 – Thermal Utilities (available on online)
3. Guide books for National Certification Examination for Energy Managers/Energy Auditors Book 3- Electrical Utilities (available on online)
4. Guide books for National Certification Examination for Energy Managers/Energy Auditors Book 4 (available on online)

Reference Books:

1. Success stories of Energy Conservation by BEE (www. Bee-india.org) Bureau of Energy Efficiency, Government of India, April 2015.
2. Utilization of electrical energy by S.C. Tripathi, Tata McGraw Hill 1991.

Code:B16EE4104

ELECTRICAL MACHINE DESIGN
(Elective-II)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students will

1. Understand the concept of magnetic circuits, temperature rise in electrical machines
2. Understand the concept of transformers design & their windings
3. Examine various losses in DC machines& their classification
4. understand the design procedures of Induction Machines & Classification
5. understand design procedures of synchronous machine and induction machines.

Course Outcomes: Students will be able to:

1. Understand the fundamental concepts of electrical machine design.
2. Design the armature, field winding and main dimensions of DC Machine.
3. Design of the single phase and three phase transformer dimensions and windings.
4. Design number of turns, phase, air gap length and conductor size etc. for given synchronous machine.
5. Find number of slots / pole and develop various winding diagrams for a given AC Machines.

SYLLABUS

FUNDAMENTAL ASPECTS OF ELECTRICAL MACHINE DESIGN: Design of Machines, Design Factors, Limitations in Design, Basic Principles,specification,Ratings, Magnetic Circuits, magnetization curves, heating, cooling, temperature rise with short term rating .(6 Hours)

D.C MACHINE: Construction Details, Armature, windings, Commutator, Design of output equation, Selection of No. of poles, Magnetic circuit and Magnetization curve.(12 Hours)

TRANSFORMER : Classification of Transformers, core construction, types of winding and design, Cooling and insulation,Output of Transformer, output equation ration of iron loss to copper loss, relation between core area and weight of iron and copper, optimum designs.(12 Hours)

THREE PHASE INDUCTION MACHINE: Stator, stator frames, rotor, rotor windings, comparison of squirrel cage and wound rotors, slip rings, design of output equation, main dimensions, stator winding, design of squirrel cage rotor and wound rotor.(12 Hours)

THREE PHASE SYNCHRONOUS MACHINE : Output equation, main dimensions for salient and non salient pole machines, armature windings and design, selection of stator slots, air gap length, design of rotor for salient pole and turbo alternators.(12 Hours)

Text Books:

1. Sawhney AK, "Electrical Machine Design", Dhanpat Rai & Sons.

Reference Books:

1. Clayton A.E., "The performance and design of D.C. Machines", Pitman (ELBS).
2. Say MG, "The performance and design of A.C. Machines", Pitman (ELBS).

Code:B16EE4105

OPERATIONS RESEARCH
(Elective-III)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students will

1. Evaluate how to formulate different objective functions applicable to transportations, Hospital management, Agriculture, Libraries, City planning, Financial Institutions, Construction Management.
2. Understand how to associate the evidential support and scientific base for Managers, Engineers and practitioners to take several decisions.
3. Remember how to relate OR methodologies for different agencies like Indian Railways, Indian Airlines, Hindustan Liver Ltd, Delhi Cloth Mills, Tata Iron & Steel co., Life Insurance Corporation of India, Fertilizer Corporation of India, etc
4. Remember how to examine different approaches to solve problems related to transportation and assignment problems.
5. Apply different logics for project management.
6. Analyse how to inspect inventory models along with Demand, production rate.

Course Outcomes:

Student are able to

1. Understand about the techniques of planning and monitoring the progress of a software project.
2. Solve problems related to Project management.
3. Analyse about the basic structure, characteristics, Functions and relationships of an organisation so that he may acquire sound, scientific and Quantitative knowledge for decision making
4. Knowledge to solve problems related to planning, transportation etc.
5. Realization of different techniques of engineering and other problems.

SYLLABUS

INTRODUCTION TO OPERATIONS RESEARCH:
Applications of OR, Optimization, Mathematical Model- Linear Programming Problem, Requirements For A LP Problem, Examples On The Application Of LPP, Graphical Solution Of 2-Variable LP Problems, General Mathematical Formulation For LPP, Canonical And Standard Forms Of LP Problem, Simplex Method, Simple Problems on Simplex Methods, Big-M Method

TRANSPORTATION PROBLEM:

Matrix Terminology, Definition and Mathematical Representation of Transportation Model, Formulation and Solution of Transportation Models (Basic Feasible Solution by North-West Corner Method, Least Cost Entry Method. Vogel's Approximation Method)

ASSIGNMENT PROBLEM:

Matrix Terminology, Definition Of Assignment Model, Comparison With Transportation Model, Mathematical Representation Of Assignment Model, Formulation And Solution Of Assignment Models.

PERT AND CPM NETWORK:

Introduction, Phases Of Project Scheduling, Network Logic, Numbering The Events (Fulkerson's Rule), Measure Of Activity, Forward Pass And Backward Pass Computations, Slack Critical Path.

GAME THEORY:

Useful Terminology, Rules For Game Theory, Saddle Point, Pure Strategy, Mini-Max, Maxi-MinPrinciple, Reduce Game By Dominance, Graphical solution, Mixed Strategies, 2x2 Games Without Saddle Point.

Text Books:

1. "Operations research-an introduction' by H.Taha, 10th Edition, Prentice Hall Of India Pvt. Ltd.
2. "Engineering Optimization-Theory & Practice" By S.S. Rao,4th Edition, New Age International (P) Ltd.

Reference Books:

1."Operations research – an introduction" by P.K.Gupta&D.S.Hira, Seventh Revised Ediotion, S.Chnd& Co. Ltd.

Code:B16EE4106

FLEXIBLE AC TRANSMISSION SYSTEMS
(Elective-III)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

1. Understand the importance of controllable parameters and benefits of FACTS controllers.
2. Know the significance of shunt, series compensation and role of FACTS devices on system control.
3. Analyse the functional operation and control of SVC, SSSC, TSSC and STATCOM.

Course Outcomes:

After completing this course students will be able to

1. Understand the needs of power systems and utility networks where installation of FACTS Controllers/Devices becomes essential.
2. Evaluate the operating principles, control systems and modelling of different FACTS Controllers.
3. Apply different FACTS controller for enhancing power transfer, enhancement of system damping, enhancing power stability and prevention of voltage instability.

SYLLABUS

INTRODUCTION: Reactive power control in electrical power transmission lines – Uncompensatedtransmission line – series compensation – Basic concepts of Static Var Compensator (SVC) – ThyristorControlled Series capacitor (TCSC) – Static Synchronous Compensator (STATCOM)- Static SynchronousSeries Compensator (SSSC).

STATIC VAR COMPENSATOR (SVC) AND APPLICATIONS: Voltage control by SVC – Advantages of slope in dynamic characteristics – Influence of SVC on systemvoltage – Design of SVC voltage regulator –Modelling of SVC for power flow and fast transient stability – Applications: Enhancement of transient stability – Steady state power transfer Enhancement of power systemdamping.

THYRISTOR CONTROLLED SERIES CAPACITOR (TCSC) AND APPLICATIONS: Operation of the TCSC – Different modes of operation – Modelling of TCSC – Variable reactance model –Modelling for Power Flow and stability studies. Applications: Improvement of the system stability limit –Enhancement of system damping.

VOLTAGE SOURCE CONVERTER BASED FACTS CONTROLLERS: Static Synchronous Compensator (STATCOM) – Principle of operation – V-I Characteristics. Applications:Steady state power transfer-enhancement of transient stability – prevention of voltage instability.

STATIC SYNCHRONOUS SERIES COMPENSATOR (SSSC): Operation of SSSC and the control of power flow – modelling of SSSC in load flow and transient stabilitystudies.

Text Books:

1. R.Mohan Mathur, Rajiv K.Varma, "Thyristor – Based Facts Controllers for Electrical TransmissionSystems", IEEE press and John Wiley & Sons, Inc, 2002.
2. Narain G. Hingorani, "Understanding FACTS -Concepts and Technology of Flexible AC TransmissionSystems", Standard Publishers Distributors, Delhi- 110 006, 2011.
3. K.R.Padiyar," FACTS Controllers in Power Transmission and Distribution", New Age International(P)Limited, Publishers, New Delhi, 2008.

Reference Books:

1. A.T. John, "Flexible A.C. Transmission Systems", Institution of Electrical and Electronic Engineers(IEEE)1999.
2. V.K. Sood, HVDC and FACTS controllers – Applications of Static Converters in Power System, APRIL2004 , Kluwer Academic Publishers, 2004.
3. Xiao – Ping Zang, Christian Rehtanz and Bikash Pal, "Flexible AC Transmission System: Modelling andControl" Springer, 2012.

Code: B16EE4107

INTRODUCTION TO SOFT COMPUTING

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives: Student will be made to:

1. Learn the various soft computing frame works
2. Be familiar with design of various neural networks
3. Be exposed to fuzzy logic
4. Learn genetic programming.

Course Outcomes: Students are able to

1. Apply various soft computing frame works like neural network, fuzzy logic and genetic algorithm.
2. Design of various neural networks and fuzzy logic modeling depending upon the required application.
3. Discuss hybrid soft computing.

SYLLABUS

Introduction: Artificial neural network: Introduction, characteristics- learning methods – taxonomy – Evolution of neural networks- basic models – important technologies – applications. Fuzzy logic: Introduction – crisp sets- fuzzy sets – crisp relations and fuzzy relations: cartesian product of relation – classical relation, fuzzy relations, tolerance and equivalence relations, non-iterative fuzzy sets. Genetic algorithm- Introduction – biological background – traditional optimization and search techniques – Genetic basic concepts.

Neural networks: McCulloch-Pitts neuron – linear separability – hebb network – supervised learning network: perceptron networks – adaptive linear neuron, multiple adaptive linear neuron, BPN, RBF, TDNN- associative memory network: auto-associative memory network, hetero-associative memory network, BAM, hopfield networks, iterative autoassociative memory network & iterative associative memory network –unsupervised learning networks: Kohonen self organizing feature maps, LVQ – CP networks, ART network.

Fuzzy Logic : Membership functions: features, fuzzification, methods of membership value assignments-Defuzzification: lambda cuts – methods – fuzzy arithmetic and fuzzy measures: fuzzy arithmetic – extension principle – fuzzy measures – measures of fuzziness -fuzzy integrals – fuzzy rule base and approximate reasoning : truth values and tables, fuzzy propositions, formation of rules-decomposition of rules, aggregation of fuzzy rules, fuzzy reasoning-fuzzy inference systems-overview of fuzzy expert system-fuzzy decision making.

Genetic Algorithm: Genetic algorithm and search space – general genetic algorithm – operators – Generational cycle – stopping condition – constraints – classification – genetic programming – multilevel optimization – real life problem- advances in GA

Hybrid Soft Computing Techniques & Applications : Neuro-fuzzy hybrid systems – genetic neuro hybrid systems – genetic fuzzy hybrid and fuzzy genetic hybrid systems – simplified fuzzy ARTMAP – Applications: A fusion approach of multispectral images with SAR, optimization of traveling salesman problem using genetic algorithm approach, soft computing based hybrid fuzzy controllers.

Text Books:

1. J.S.R.Jang, C.T. Sun and E.Mizutani, "Neuro-Fuzzy and Soft Computing", PHI / Pearson Education 2004.
2. S.N.Sivanandam and S.N.Deepa, "Principles of Soft Computing", Wiley India Pvt Ltd, 2011.

Reference Books:

1. S.Rajasekaran and G.A.Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis & Applications", Prentice-Hall of India Pvt. Ltd., 2006.
2. George J. Klir, Ute St. Clair, Bo Yuan, "Fuzzy Set Theory: Foundations and Applications" Prentice Hall, 1997.
3. David E. Goldberg, "Genetic Algorithm in Search Optimization and Machine Learning" Pearson Education India, 2013.
4. James A. Freeman, David M. Skapura, "Neural Networks Algorithms, Applications, and Programming Techniques, Pearson Education India, 1991.
5. Simon Haykin, "Neural Networks Comprehensive Foundation" Second Edition, Pearson Education, 2005.

Code:B16EE4108

MICROPROCESSOR AND MICRO CONTROLLER LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

Student will

1. Learn about working of 8085 microprocessor and instructions sets.
2. Aquire knowledge of writing program for artimetic, logical operation to perform required task like pick largest even number, 8-bit array addition etc.
3. Understand the programs to display decimal count & converting binary to BCD codes and also learns about sorting techniques.
4. Learn about working of 8051 microcontroller and instructions sets
5. Learn about basic programming skills using 8051 contoller and also acquire knowledge on interfacing with other devices like traffic light controller, stepper motor etc

Course Outcomes:

Students are able to

1. Evaluate the programs using basic fundamentals of 8085 Microprocessor& 8051 Microcontroller.
2. Develop different programs on extended version like 8086 microprocessor.
3. Design programs for interfacing circuits like traffic controller, LED display board, Motor controllers etc.

LIST OF EXPERIMENTS

PART A: EXPERIMENTS ON MICROPROCESSORS:

1. Program to add two 8-bit binary numbers
2. Program to add an array of 8-bit binary numbers.
3. Program to pick the largest even number from an array of 8- bit binary numbers
4. Program to find the sum of an array of 2- digit packed BCD numbers.
5. Program to display decimal count from 0 to 9 with suitable delay between each count.
6. Program to convert an 8- bit binary number into BCD.
7. Program to sort given array of 8-bit binary numbers.

PART B: EXPERIMENTS ON MICRO CONTROLLERS:

8. Microcontroller programming on two 8-bit numbers
   a)Addition, b) Subtraction , c) Multiplication & d) Division
9. Program to obtain decimal equivalent of an 8-bit hexadecimal number
10. Interfacing stepper motor and speed control using 8051 microcontroller
11. Traffic light control using 8051 microcontroller.

Text Books:

1. R.S. Gaonkar: Microprocessor Architecture.
2. Microprocessors & Its Applications ByTheagarajan, R., Dhanpal, S. &Dhanaseturan, S., New Age India Ltd., 1998
3. A.K.Ray and Bhurchandi, “Advanced Microprocessors and Peripherals”, 2nd Edition, TMH Publications
4. Ajay V. Deshmukh, “Microcontrollers, Theory and applications”, Tata McGraw-Hill Companies – 2005.

Reference Books:

1. Douglas V.Hall, “Microprocessors and Interfacing”, 2nd Revised Edition, TMH Publications.
   Design”, 2nd ed., PHI
2. Kenneth. J.Ayala, The 8051 Microcontroller Architecture, Programming and Applications, Penram International 2nd edition, 1996.
3. Krishna Kant, “Microprocessors and Microcontrollers”, PHI Publishers

Code:B16EE4109

POWER ELECTRONICS LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

After Completion of the course student will be able to

1. Evaluate the effect of Freewheeling diode on single-phase semi converter with RL-load & single-phase full converter with DC motor.
2. Determine the effect of frequency on series inverter & Mc-Murray Bedford Inverter.
3. Investigate the operating modes of chopper in constant frequency & constant duty cycle.
4. Analyze the AC output voltage of AC voltage controller &Cycloconverter with changing frequency modes.
5. Investigate the SCR triggering using R, RC & UJT methods.

Course Outcomes:

Students are able to

1. Analyze the Thyristor and Transistor characteristics.
2. Evaluate the characteristics of converter and inverter with R and RL Loads.
3. Apply different controllers for controlling the DC drives

LIST OF EXPERIMENTS

1. To vary dc output voltage of a single phase semi converter with R-L load.
2. To vary D.C output voltage using impulse commutated chopper.
3. To obtain the ac output voltage of a resonant series inverter at different frequencies.
4. To operate UJT as relaxation oscillator. obtain the characteristics of UJT
5. To vary the firing angle of thyristor using R & RC Triggering
6. To vary the speed of dc motor by using single phase full converter with and without free wheeling diode.
7. To find performance parameters of uncontrolled rectifiers (center tapped, Bridge type).
8. To vary single phase AC voltage-using single phase AC voltage controller.
9. To vary frequency of single phase AC voltage using 1-Φ to 1-Φmid pointcyclo converter.
10. To vary frequency of mcmurraybedford inverter
11. To vary D.C output voltage using impulse commutated chopper.

Text Books:

1. Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition
2. Power electronics – M.D. Singh & K.B. Kanchandhani, Tata Mc Graw – Hill Publishing Company, 2nd edition.

Reference Books:

1. Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3 rd edition.
2. Power Electronics – VedamSubramanyam, New Age International (p) Limited, Publishers.
3. Power Electronics – P.C. Sen, Tata Mc Graw-Hill Publishing.
4. Thyristorised power Controllers – G.K. Dubey, S.R Doradra, A. Joshi and R.M.K. Sinha, New Age international Pvt Ltd. Publishers latest edition

Code: B16EE4110

PROJECT PHASE-I

Lab	**: 3 Periods**	**Sessionals**	**: 50**
		Credits	**: 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

IV/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards**)**
Under Choice Based Credit System

ELECTRICAL AND ELECTRONICS ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Sessional Marks	Exam Marks	Total Marks
B16 EE 4201	Power System Operation & Control	4	3	1	-	4	30	70	100
# ELE-IV	ELECTIVE-IV	4	3	1	-	4	30	70	100
B16 EE 4205	Simulation Lab	2	-	-	3	3	50	50	100
B16 EE 4206	Project Phase-II	12	-	-	9	9	50	100	150
	Total	**22**	**6**	**2**	**12**	**20**	**160**	**290**	**450**

ELECTIVE-IV	B16 EE 4202	Advanced Power Electronics
	B16 EE 4203	Electrical Distribution Systems
	B16 EE 4204	High Voltage Direct Current **(**HVDC)

Code:B16EE4201

POWER SYSTEM OPERATION & CONTROL

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students will

1. Understand the concepts of economical operation using thermal plants and also combined hydro- thermal plants.
2. Study the optimal unit commitment problem and solution.
3. Understand the optimal power flow problem and solution.
4. Study the Single area and Two area system under steady state response and dynamic of both controlled and uncontrolled case study.
5. Understand The concepts of controlling frequency and voltage using ALFC & AVR.
6. Study the contingency analysis and power system security
7. Study the stability enhancement methods & preventive control.

Course Outcomes:

Students are able to

1. Understand in solving economic load scheduling and unit commitment problems using various computational methods for optimal operation of generators.
2. Solve the optimal scheduling of Thermal and Hydro-thermal system and Optimal power flow.
3. Analyze the concepts like Load Frequency Control, Power system and stability issues.

SYLLABUS

OPTIMAL SYSTEM OPERATION:

Optimal operation of Generators in Thermal power stations, Heat rate curve, Cost Curve, Incremental fuel and Production costs, Input–output characteristics. Optimum generation allocation with & without transmission line losses, Loss Coefficients, General transmission line loss formula. Optimal scheduling of Hydrothermal System**:** Short term hydrothermal scheduling problem.

UNIT COMMITMENT & OPTIMAL POWER FLOW

Optimal unit commitment problem: Need for unit commitment, Constraints in unit commitment, Cost function formulation, Solution methods using Priority list method & Dynamic programming. Optimal Power Flow: Problem formulation & Solution of OPF by Gradient Method.

AUTOMATIC GENERATION CONTROL:

Frequency control: Load-Frequency Control Concepts, Load frequency Control of a Single Area System modelling, Steady state & Dynamic response of uncontrolled & controlled cases, Two-area system modelling - Static analysis of uncontrolled case, Tie line with frequency bias control of two-area system.**Voltage control:** Automatic voltage regulator, Exciter types, Exciter modelling, Generator modelling, static and dynamic response of AVR loop.

POWER SYSTEM SECURITY

Introduction, Factors affecting the power system security, contingency analysis procedures, linear sensitivity factors, line outages distribution factors and genearation shift factors and its derivation, AC power flow method, contingency selection.

EMERGENCY CONTROL:

Concepts, Preventive and Emergency Control, Coherent Area Dynamics, Stability Enhancement Methods, Long Term Frequency Dynamics, Average System Frequency, Centre of Inertia.

Text Books:

1. Power System Engineering By I.G. Nagarath& D.P. Kothari, Fouth edition, Tata Mc Graw Hill Publications
2. Electric Energy Systems Theory-An Introduction ByOlle I. Elgerd, second edition Tata Mc Graw Hill publication.
3. Power system operation and control byP.S.R.Murthy, second edition B.S Publication.

Reference Books:

1. Advanced power system operation and control by Allen.J.wood and B.F. Wollenberg, second edition, John wiley& sonspulication Inc. 1984.
2. Advanced Power System Analysis and Dynamics By L.P. Singh, Third Edition, Wiley Eastern Limited publications.
3. Power System Analysis By HadiSadat,Third Edition,Tata Mc Graw Hill publication.

Code: B16EE4202

ADVANCED POWER ELECTRONICS
(Elective-IV)

Theory	**: 3 Periods**	**Sessionals**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives: Students will

1. Understand the characteristics and principle of operation of modern power semiconductor devices.
2. Comprehend the concepts of different power converters and their applications.
3. Analyze and design switched mode regulator for various applications.
4. Describe the importance of AC voltage controllers and various converters for various industrial applications

Course Outcomes: Upon the completion of this course, the student will be able to

1. Choose appropriate device for a particular converter topology.
2. Understand the operating principles and models of different types of power electronic converters including dc-dc converters, AC to AC converters and inverters
3. Analyze various converter topologies and can identify the corresponding T.H.D.
4. Apply the knowledge of various devices and converter topologies for different industrial applications.

SYLLABUS

MODERN POWER SEMICONDUCTOR DEVICES Modern power semiconductor devices – MOS turn Off Thyristor (MTO) – Emitter Turn off Thyristor (ETO) – Integrated Gate-Commutated thyristor (IGCTs) – MOS-controlled thyristors (MCTs) – Static Induction circuit – comparison of their features

D.C. TO D.C. CONVERTER Classification of choppers. Principle of operation, steady state analysis of class A chopper, step up chopper, switching mode regulators: Buck, Boost, Buck-Boost, Cuk regulators. Current commutated and voltage commutated chopper. Power Supplies: Switched mode D.C. and A.C. power supplies. Resonant D.C and A.C. power supplies.

A.C. TO A.C. CONVERTER Classification, principle of operation of step up and step down Cycloconverter. Single phase to single phase Cycloconverter with resistive and inductive load. Three phase to single phase Cycloconverter: Half wave and full wave. Cosine wave crossing technique. Three phases to three phase Cycloconverter. Output voltage equation of Cycloconverter.

D.C. TO A.C. CONVERTER Classification, basic series and improved series inverter, parallel inverter, single phase voltage source inverter, steady state analysis, Half bridge and full bridge inverter: Modified Mcmurray and Modified Mcmurray Bedford inverter, voltage control in single phase inverters, PWM inverter, reduction of harmonics, current source inverter, three phase bridge inverters.

MULTILVEL CONVERTERS AND PWM TECHNIQUES: Introduction to multilevel converters – diode clamp, flying capacitor and cascaded-cell converters;PWM with a few switching angles per quarter cycle, equal voltage contours, selective harmonic elimination, THD optimized PWM, off-line PWM, Phase disposition techniques.

Text Books:

1. Power Electronics: Circuits, devices and applications - M.H. Rashid- PHI., Third Edition.
2. Power Electronics: Converters, Applications and Design - Ned Mohan, Tore M. Undeland, William P. Robbins - John Wiley & Sons., Third Edition.

Reference Books:

1. Power Electronics - P.S. Bimbhra, Khanna Publishers., Fifth Edition.
2. An Introduction to Thyristors and their applications - M. Ramamoorthy -East-West Press., Second Edition.
3. Power Electronics- M.D. Singh and K.B. Khanchandani, Tata McGraw-Hill., Second Edition.

Code:B16EE4203

ELECTRICAL DISTRIBUTION SYSTEMS
(Elective-IV)

Theory	**: 3 Periods**	**Sessional**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students will

1. Study different factors of Distribution system.
2. Study and design the substations and distribution systems.
3. Study the concepts of voltage drop and power loss.
4. Study the distribution system protection and its coordination.
5. Study the effect of compensation for power factor improvement and effect of voltage control on distribution system.

Course Outcomes:

Students are able to

1. Understand various concepts of distribution system, protection and its coordination.
2. Apply the best methods for power factor improvement and voltage control.
3. Differentiate the types of loads and their characteristics .

SYLLABUS

GENERAL CONCEPTS:

Introduction to distribution systems, Load modelling and characteristics – Coincidence factor– Contribution factor loss factor – Relationship between the load factor and loss factor – Classification of loads (Residential, commercial, Agricultural and Industrial).

SUBSTATIONS:

Location of substations**:** Rating of distribution substation – Service area with‘n’primaryfeeders – Benefits and methods of optimal location of substations..

DISTRIBUTION FEEDERS:

Design Considerations of distribution feeders**:** Radial and loop types of primary feeders –Voltage levels – Feeder loading – Basic design practice of the secondary distribution system.

SYSTEM ANALYSIS:

Voltage drop and power–loss calculations**:** Derivation for voltage drop and power loss inlines – Uniformly distributed loads and non-uniformly distributed loads – Numericalproblems - Three phase balanced primary lines.

PROTECTION:
Objectives of distribution system protection – Types of common faults and procedure forfault calculations for distribution system – Protective devices**:** Principle of operation of fuses– Circuit reclosures – Line sectionalizes and circuit breakers.

COORDINATION:
Coordination of protective devices**:** General coordination procedure –Various types of coordinated operation of protective devices - Residual Current Circuit Breaker

COMPENSATION FOR POWER FACTOR IMPROVEMENT:
Capacitive compensation for power factor control – Different types of power capacitors –shunt and series capacitors – Effect of shunt capacitors (Fixed and switched) – Power factorcorrection – Capacitor allocation – Economic justification – Procedure to determine the bestcapacitor location – Numerical problems.

VOLTAGE CONTROL:
Voltage Control**:** Equipment for voltage control – Effect of series capacitors – Effect ofAVB/AVR – Line drop compensation – Numerical problems.

Text Books:

a. “Electric Power Distribution system, Engineering” - 3rd Edition by Turan Gonen, CRC Press,2015.

Reference Books:

1. “Electrical Distribution Systems”- 2nd Edition by Dale R.Patrick and Stephen W.Fardo,CRC press,2009.
2. “Electric Power Distribution” - 4th Edition by A.S. Pabla, Tata McGraw–hill Publishingcompany,1997.
3. “Electrical Power Distribution Systems”- 1st Edition by V.Kamaraju, Right Publishers,2009.

Code:B16EE4204

HIGH VOLTAGE DIRECT CURRENT TRANSMISSION
(Elective-IV)

Theory	**: 3 Periods**	**Sessional**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 4**

Course Objectives:

Students will

1. Learn about the concept of HVDC transmission
2. Know about types of Harmonics produced by converters in HVDC systems
3. Learn control techniques of converters used in HVDC transmission systems
4. Know interaction between HVAC and HVDC transmission systems

Course Outcomes:

Students are able to

1. Understand about the importance of HVDC transmission over HVAC transmission.
2. Analyze different types of Harmonics produced by converters in HVDC systems
3. Apply different types of controlling techniques for converters used in HVDC systems
4. Describe interaction between HVAC and HVDC transmission systems

SYLLABUS

H.V.D.C. TRANSMISSION: General considerations, Power Handling Capabilities of HVDC Lines, Basic Conversion principles, static converter configuration.

STATIC POWER CONVERTERS: 3-pulse, 6-pulse and 12-pulse converters, converter station and Terminal equipment, commutation process, Rectifier and inverter operation, equivalent circuit for converter – special features of converter transformers.

Harmonics in HVDC Systems, Harmonic elimination, AC and DC filters.

CONTROL OF HVDC CONVERTERS AND SYSTEMS: constant current, constant extinction angle and constant Ignition angle control. Individual phase control and equidistant firing angle control, DC power flow control.

Interaction between HV AC and DC systems–Voltage interaction, Harmonic instability problems and DC power modulation.

Text Books:

1. K.R.Padiyar : High Voltage Direct current Transmission, Wiley Eastern Ltd 2ND Edition
2. E.W. Kimbark : Direct current Transmission, Wiley Inter Science Volume-2ND Edition

Reference Books:

1. Arillaga : H.V.D.C.Transmission Peter Peregrinus ltd., London UK 1983
2. E.Uhlman : Power Transmission by Direct Current, Springer Verlag, Berlin Helberg –1985.

Code:B16EE4205

SIMULATION LAB

Lab	**: 3 Periods**	**Sessionals**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

Students will

1. Learn the time domain response for the various networks and load flows of the power systems by using Matlab programming software.
2. Obtain the load frequency response and other responses of the power systems by using Simulink software

Course Outcomes:

At the end of course the student will be able to:

1. Analyze matlab program for the Ybus, load flows and Economic Load Dispatch considering with and without losses.
2. Design the Simulink models for the simulation of transient and steady state stabilities in power systems, automatic voltage regulator, load frequency control of single and two area system.

LIST OF EXPERIMENTS

1. Series Rlc Circuit
2. Formation Of Y-Bus & Z-Bus
3. Solution Of Power Flow Using Gauss Siedel Method.
4. Abcd Constants For Long Lines And Voltage Profile Observation For Open Circuit Line With And With Out Shunt Reactor Compensation
5. Optimal Operation Of Generating Units Without Considering Transmission Losses [Economic Load Dispatch]
6. Optimal Generator Scheduling Considering Transmission Losses
7. Simulation Of Steady State Stability For Small Disturbances With & Without Change In Power Input
8. Simulation Of Transient Stability
9. Simulation Of Automatic Voltage Regulator Using Both Stabilizer And Pid Controller.
10. Simulation Of Load Frequency Control Of A Single Area System With Free Governor Operation
11. Simulation Of Load Frequency Control Of A Two Area System With Free Governor Operation
12. Load Frequency Control Of A Two Area System With Tie Line Biased Control

Text Books:

1. Power System Engineering By I.G. Nagarath& D.P. Kothari, Fouth edition, Tata Mc Graw Hill Publications
2. Electric Energy Systems Theory-An Introduction ByOlle I. Elgerd, second edition Tata Mc Graw Hill publication.
3. Power system operation and control by P.S.R.Murthy, second edition B.S Publication.

Reference Books:

1. Advanced power system operation and control by Allen.J.wood and B.F. Wollenberg, second edition, John wiley& sons pulication Inc. 1984.
2. Advanced Power System Analysis and Dynamics By L.P. Singh, Third Edition, Wiley Eastern Limited publications.
3. Power System Analysis By HadiSadat,Third Edition, Tata Mc Graw Hill publication.

Code: B16EE4206

PROJECT PHASE-II

LAB : 9 Periods	**Sessionals**	**: 50**
	Ext. Marks	**: 100**
	Credits	**: 12**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation.
External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:

1. Cover Page & Title Page .
2. Bonafide Certificate
3. Abstract.
4. Table of Contents
5. List of Tables
6. List of Figures
7. List of Symbols, Abbreviations and Nomenclature
8. Chapters
9. Appendices
10. References

*The table and figures shall be introduced in the appropriate places.